

Process Control

Final Lab Report

Unit Operations Lab 1

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Group 3

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1. Executive Summary

This lab was divided into three separate experiments, with the main objective of each to plot the water level versus time for different settings and to calculate the offset for each setting used in the experiment using the Armfield PCT50 Level Control. Experiment A utilized the manual control of the open loop, Experiment B uses only the proportional band setting of the PID control on a closed loop, and Experiment C utilizes both the proportional band and integral time settings of the PID control on a closed loop. The raw data of each experiment is recorded using the built-in data recording software, and the calculated values for each respective experiment are summarized below.

Experiment B	
Proportional Band Value, %	Average Offset
2	2.58
5	-0.03
10	-3.63
20	-9.49
50	-25.19
100	-51.9
200	30.31

Table 1: Summary of Calculated Values of Experiment B

Experiment C	
Integral Time Value, s	Average Offset
5	-0.72
25	-1.38
50	0.20
100	-2.43

Table 2: Summary of Calculated Values of Experiment C

From tables 1 and 2, it's observed that the higher the proportional band value, the lower the offset, except for 200%. Additionally, as the integral time value increases, the offset decreases, with the outlier being 50s.

2. Introduction

The objective of this lab was to plot the water level against time under different settings and to calculate the offset that correlates with these settings. Experiments involving process control unit are especially important to chemical engineers, since level control units such as the Armfield PCT50 are widely used in the petroleum and brewing industries to ensure proper operating conditions as well as adequate solution level to prevent accidents [1]. For all three experiments, a computer was used as a way to control the pump as well as to read output, an Armfield plastic vessel was used as a container for the experiment to proceed, a drain valve allowed the vessel to be drained, a solenoid

valve was used to control the flow of water to and from the upper process vessel, a water pump was used to transfer the water throughout the vessel, and a pressure sensor was used to read the pressure of water flowing out of upper process vessel. To prepare for experiment 1, the PCT50 software was started and “experiment 1” was selected. The tank was then filled up to the fill line with deionized water, and CV2 valve was fully opened. After the CV1 valve was checked for proper operation, it was fully opened, and the pump speed was set to 50%. To perform the “initial level control” of experiment 1, the pump speed was slowly increased, and the steady-state pump speed was recorded before the “GO” icon was used for data collection. With the data collection still happening, the “Disturbance” section of experiment 1 was preceded by first opening the solenoid valve, to which the water level was allowed to fall about 20 mm before the pump speed was increased to allow the water to reach its original level. The solenoid valve was then closed, and the water level was allowed to rise for about 20mm. The pump speed then decreased until the water level once again reached its original level. Moving onward to the “water fill and drain rate” of experiment 1, a new results sheet was first created, and then the pump speed was gradually increased until it’s greater than 75%. After allowing the water level to increase to 100mm, the pump was stopped and the vessel was allowed to drain. Once the vessel was empty, the pump speed was set to 75% and the water level was allowed to reach 100mm once again. Then pump was then stop and the vessel was allowed to drain once more. Proceeding to experiment 2, the potential band P was set to 200%, integral time I to 0, derivative time D to 0 within the PID box, and the water level was set to 75mm. After the automatic mode of operation was chosen, the water level was allowed to reach approximately 75mm. The solenoid valve was then opened for disturbance and then closed after the water level had reached somewhat steady-state. The setpoint was then adjusted to 100mm, allowed to reach steady-state, and then returned to 75mm. The P value was then set to 100%, 50%, 20%, 10%, 5%, and 2%, with each value following an identical procedure as the one described for 200%. For experiment 3, the proportional band P was set to 20%, integral time I was set to 0, and the derivative time D to 0. The set point was set to 75mm, and automatic mode of operation was chosen before the water level was allowed to increase. Upon the water level reaching steady-state, I was then set to 100. The solenoid valve was then opened to disturb the process and closed upon the water level reaching 75mm. The water was then set to 100mm, and its behavior was observed before the level was set back to 75mm. Starting at I value of 50, different I values were arbitrarily chosen for the trials, which they follow an identical procedure as described above, per [2].

3. **Theory**

In this experiment, the Armfield PCT50 level unit is adapted as a single-input/output control system. Here, the controlled variable is the tank liquid level (measured by an electronic pressure sensor), and the manipulated variable is the pump speed, which directly changes the inlet flow, . The dynamics follow an unsteady mass balance;

therefore, any mismatch between inflow and outflow causes the level to rise or fall, and the level processes exhibit first-order behavior when linearized about an operating point. However, the outflow, Q , is set by an outlet restriction that can be affected by outside changes, e.g. opening a solenoid valve to increase the outflow. This can be done in two operations: open loop, where the operator must adjust the pump speed to restore the balance (steady-state) after disturbances/changes; or closed loop, where a controller automatically adjusts the pump speed from an error e which drives the level to the setpoint and attempts to reject these disturbances/changes. These definitions/relationships follow standard process control treatment, as summarized in Perry's Chemical Engineers' Handbook [3]. A summary of variables and their corresponding units is provided below.

Table 2: Variables and Their Description

Variable	Units	Description
h	mm	Controlled variable – measured tank level
	mm	Desired target level
$e(t)$	mm	Control error signal used by the controller
N	% OR rpm	Manipulated variable – pump drive command that changes inlet flow
	OR	Flow into the tank
	OR	Flow leaving the tank
P	kPa OR Pa	Pressure at the sensor used to infer liquid level
t	s	Time variable for dynamic response

4. Results & Data Analysis

a. Experiment 1

The first disturbance pump speed level disturbance

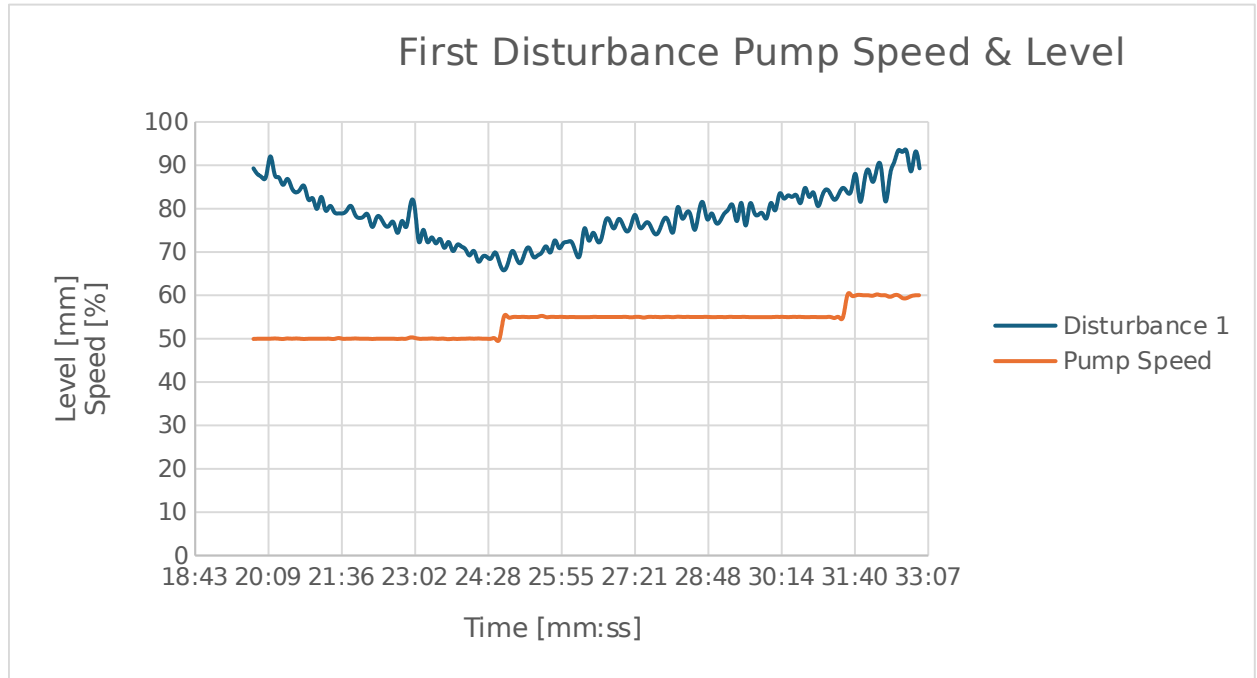


Figure 1: First disturbance with pump speed and level of the tank

At the start of the first disturbance the tank is near 90mm. When the solenoid valve is opened there is another outflow path that is introduced. This causes the outflow to be greater than the inflow rate. Because of this the water level begins to decrease gradually falling to around 66-70mm. During the first ~7 minutes the pump speed stays at 50% which means inflow is not increasing enough to counter the outflow. Halfway through the run the pump increased to 55% and later to 60%. The step increases are attempts to raise the inflow rate so that it can exceed the outflow rate. As a result, the tank level slowly recovers back to its original value of 90mm. As seen through the trend the slight speed pump adjustments leads to the inflow to gradually catching up and eventually exceeding the outflow which leads to the recovery level.

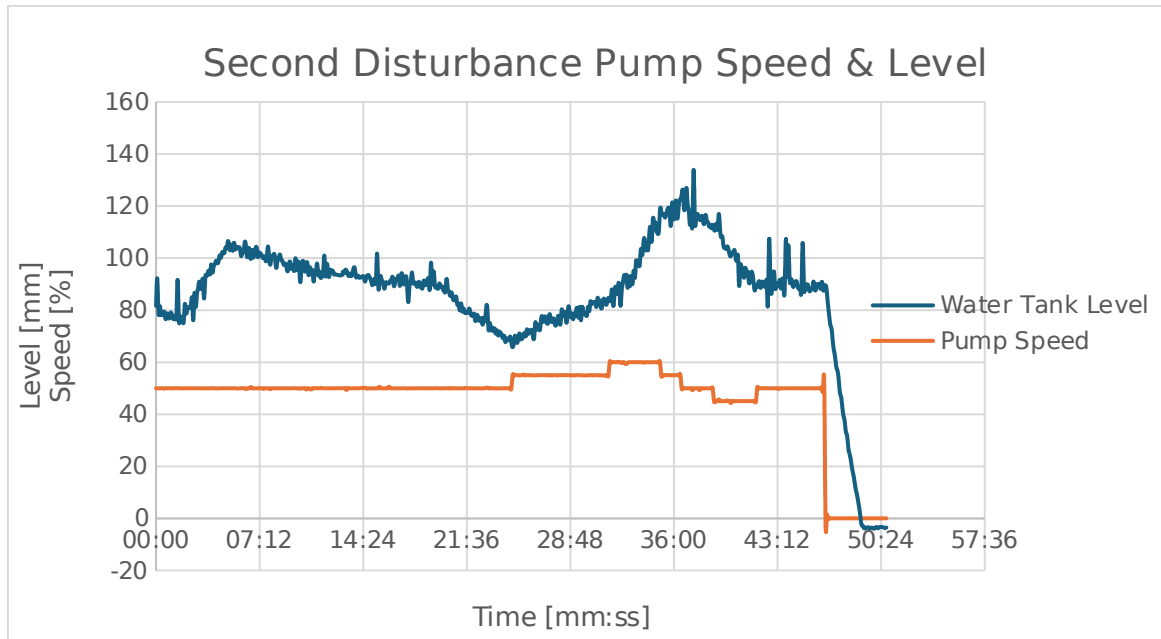


Figure 2: Second disturbance with pump speed and level of the tank

In the second graph the level initially rises to 110-115mm due to the solenoid valve being closed this causes the inflow to be greater than the outflow. Then proceeding to open the solenoid valve it creates another outflow path. As it trends downward there is noticeable noise in the middle of the runs. This causes the level in the tank to decrease gradually. Letting it drop to around 70mm then proceeding to increase the pump speed to increase the inflow to match and eventually exceed the outflow. From here the pump is reduced in small steps from 60% to about 40% pump speed, then later returning to 50% pump speed. At the end of the experiment the pump speed drops to 0%. This causes a significant decrease in the tank level. This responds exactly how it should be expected.

b. Experiment 2

All graphs included in appendix

Offset analysis

The first P value under investigation was 200%. As expected, using such a high P value resulted in a large offset from the set point throughout operation:

Table 3: Offset for disturbances at P=200%

P 200%	Valve Position	Level	Set Point	Offset
1	Closed	61.00	75.00	-14.00
2	Open	130.25	75.00	55.25
3	Closed	80.56	75.00	5.56
4	Open	77.17	75.00	-23.36
5	Closed	51.64	75.00	-23.36
6	Open	150.00	75.00	73.61
7	Closed	149.88	100.00	49.88
8	Closed	150.00	75.00	75.00
9	Open	149.17	75.00	74.17
Avg Tot				30.31

For this table averages were taken from the steady values over a length of time to get comparable results. For this trial the average offset appears to be a lower value of 30.31 but that does not consider the wild fluctuations between positive and negative values. This fluctuation was not present in other trials and suggest that a value of 200% is too wide of a tolerance and cannot properly control the liquid height. This is seen with the liquid level in the tank reaching a height of 150.0 that then began to overflow back down into the reservoir. This heavily suggested that such a high P value resulted in an unresponsive system leading to the flooding witnessed in lab. It stands to reason that reducing the P value should result in a smaller offset and is seen for the P value of 100% as pictured in table 4 below:

Table 4: Offset for disturbances at P=100%

P 100%	Valve Position	Level	Set Point	Offset
1	Open	9.5	75.0	-65.5
2	Closed	27.4	75.0	-47.6
3	Open	17.7	75.0	-57.3
4	Closed	28.3	75.0	-46.7
5	Open	27.6	100.0	-47.4
6	Closed	44.0	75.0	-47.0
Total Avg				-51.9

This run showcases when the tank is filled and reaches steady state marked as #2 in the table alongside the opening of the solenoid valve as #3. The valve is then closed and allowed to reach steady state which is marked as #4 above and then the set point is increased to 100 as marked in #5. This labeling convention was repeated for all trials for each P value. As the tank filled and reached a steady state there was an extremely large offset value of -65.5. Such a large value suggests that the controller was unable to bring the liquid level in the tank close to the set point

using only a P system at this high of a value. Reducing the P value did tend to decrease the offset as is seen in table 5:

Table 5: Offset values for disturbances at P=50%

P 50%	Valve Posi	Level	Set Point	Offset
1	Closed	46.74	75	-28.26
2	Open	51.23	75	-23.77
3	Closed	74.33	100	-25.67
4	Closed	53.72	75	-23.07
Total Avg				-25.19

For a band value of 50% there was a initial level steady offset of -28.26 upon filling the tank with similar values obtained for the first disturbance with the opening of the solenoid valve. The value offset value once again dropped when the solenoid opened and reached the lowest value pictured in the table above when the set-point was set back to 75. These values are lower than for the 100% and 200% but the lower band values give lower offset values as is shown in table 6:

Table 6: Offset for disturbances at P=20%

P 20%	Valve Posi	Level	Set Point	Offset
1	Closed	64.60	75.00	-10.40
2	Open	64.33	75.00	-10.67
3	Closed	66.00	75.00	-9.00
4	Closed	89.46	100.00	-10.54
5	Closed	68.16	75.00	-6.84
Total Avg				-9.49

As is expected, reducing the band value reduced to offset values once again. With an closed solenoid the offset values are nearly a third of what they were at the 50% band value. The largest contrast in values is the initial filling of the tank suggesting that a lower band value fills the tank more accurately. The overall offset is -9.49, which is much lower than the -25.19 for the 50% band value. This further supports the idea that reducing the band value should bring the steady value closer to the set point. This trend is further observed in the table below:

Table 7: Offset point for disturbances at P=10%

P 10%	Valve Position	Level	Set point	Offset
1	Closed	73.50	75	-1.50
2	Open	71.83	75	-3.17
3	Closed	69.67	75	-5.33
4	Closed	90.71	100.00	-9.29
5	Closed	76.13	75.00	1.13
Total Avg				-3.63

Once again, reducing the band value resulted in a lower offset value with this time an average value of -3.63 being achieved. This trend continues with the lowest band setting of 5% giving the lowest offsets at each disturbance as pictured below in table 7:

Table 8: Offset vs disturbances at P=5%

P 5%	Valve Position	Level	Set Point	Offset
1	Closed	76.33	75	1.33
2	Open	74.80	75	-0.20
3	Closed	70.00	75	-5.00
4	Closed	98.95	100.00	-1.05
5	Closed	79.79	75.00	4.79
Total Avg				-0.03

Of the tested values the 5% band value produced the lowest average offset with a value of -0.03. This follows the predicted trend and suggests that this is the optimal operational value as it gives the closest to the desired set-point values. The 2% band values resulted in a higher average value as can be seen in table 8 below:

Table 9: Offsets vs band values for P=2%

P 2%	Valve Position	Level	Set Point	Offset
1	Closed	77.42	75	2.42
2	Open	75.90	75	0.90
3	Closed	76.00	75	1.00
4	Closed	100.3	100	0.25
5	Closed	83.31	75.00	8.31
Total Avg				2.58

The average offset for 2% being higher than 5% suggests that a 2% band is too aggressive while values larger than 5% are too loose to provide optimal control of the system. It is for these reasons that 5% was calculated to be the best for the P control only. Integrating in integral time will give a more dynamic and accurate control system as is explored in experiment 3.

c. Experiment 3

Table 10: Offset for Integral Time of 100s

I 100s	Valve Position	Level	Set Point	Offset
1	Closed	68.50	75.00	-6.50
2	Open	73.29	75.00	-1.71
3	Closed	74.54	75.00	-0.46
4	Closed	96.09	100.00	-3.91
5	Closed	75.40	75.00	0.40
Total Avg				-2.43

As the integral time was adjusted for experiment 3 the main difference noticed in the graphs was how the controller was able to adjust to the disturbance introduced by the valve. In this experiment the longest setting was [I = 100s] and gave the worst performance. With the initial level control having -6.5 mm having the worst offset with the controller picking up some slack

with the later trials but overall it can be concluded to be the least effective at controlling the level in the vessel.

Table 12: Offset for Integral Time of 50s

I 50s	Valve Posi	Level	Set Point	Offset
1	Closed	72.56	75.00	-2.44
2	Open	76.08	75.00	1.08
3	Closed	80.00	75.00	5.00
4	Closed	99.25	100.00	-0.75
5	Closed	73.10	75.00	-1.90
Total Avg				0.20

When the integral time was reduced to $[I=50s]$ the offsets became much more reasonable and the system was able to achieved much closer offsets overall. The offsets calculated for this trial show that the controller was slightly less consistent in its targeting some times over shooting or undershooting which contributes to the total average of offsets being so low in comparison to the other trials.

Table 13: Offset for Integral Time of 25s

I 25 s	Valve Posi	Level	Set Point	Offset
1	Closed	71.67	75.00	-3.33
2	Open	75.00	75.00	0.00
3	Closed	77.00	75.00	2.00
4	Closed	97.07	100.00	-2.93
5	Closed	72.38	75.00	-2.63
Total Avg				-1.38

For $[I=25s]$ the controller was able to hold the system near its set point but the overall offsets leaned more negative. The disturbance section where the secondary out flow valve was held open shows a perfect offset of zero over the period. The largest mishap the controller had was for the 100mm set point where the averaged value came to $\sim 97\text{mm}$ for an offset of -3mm . Overall this is an improvement over the 100s integral time but not centered around the target as for the 50s integral time.

Table 14: Offset for Integral Time of 5s

I 5s	Valve Posi	Level	Set Point	Offset
1	Closed	71.67	75.00	-3.33
2	Open	74.14	75.00	-0.86
3	Closed	77.50	75.00	2.50
4	Closed	100.15	100.00	0.15
5	Closed	72.9231	75.00	-2.08
Total Avg				-0.72

For when the integral time is $[5s]$ the offsets stayed small but when looking at the graph the controller looks to be overreacting to disturbances with. The controller performed really well at the 100mm setpoint and when the disturbance was enabled but struggled with maintaining the 75mm setpoint.

Over these four trials the data shows that 100s is too slow of a response time and can not adapt to the change in the system. The best data is for 50s since it had the lowest overall offset. And for 25s and 5s the controller can hold the level close to the setpoint but also shows more instability in its dataset.

5. Discussion

In experiment 1, the rate at which the water drains relies heavily on the pump speed, which is an expected behavior. This is because as the pump speed increases, more water is introduced into the upper vessel, adding water weight. And as more water is introduced, the weight on top of the drain hole increases, which forces the water to rush out more quickly. To put this behavior into numerical perspective, if the pump flow rate is 30 , the upper process vessel would have about 30g of water every minute per [4].

Comparing that water amount to a flow rate of 60 , which equates to around 60 g of water

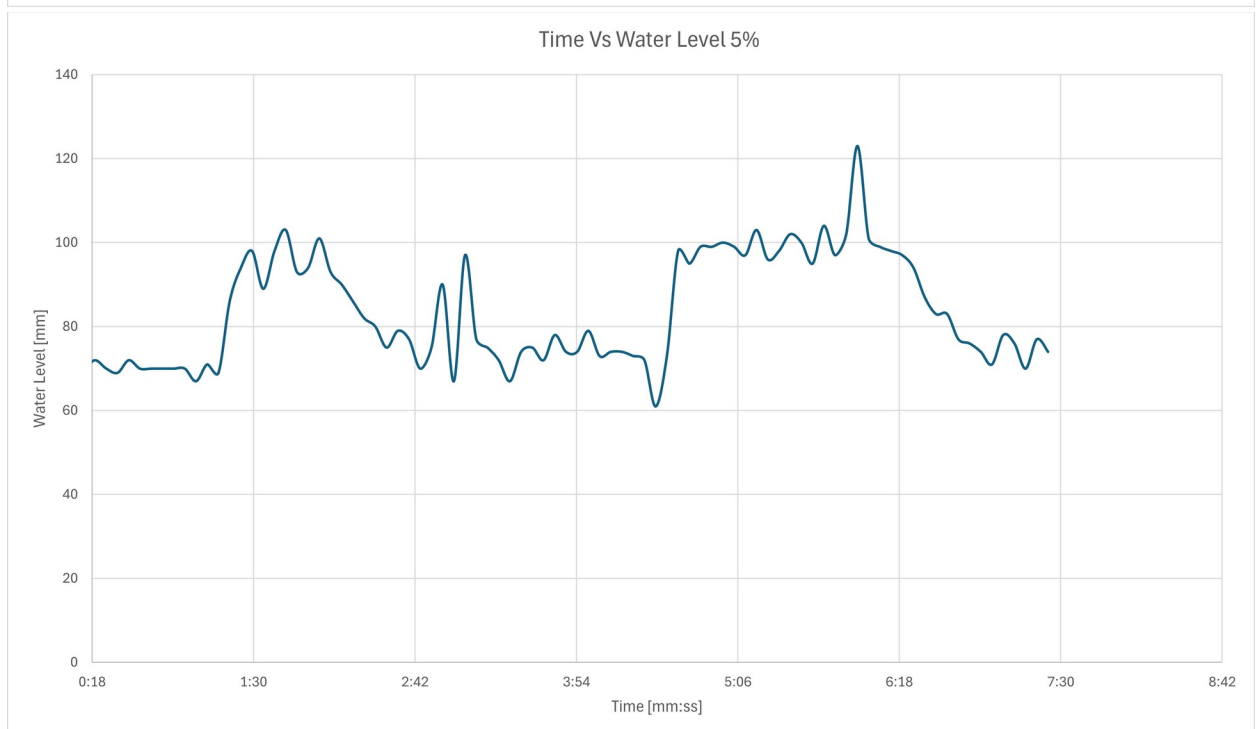
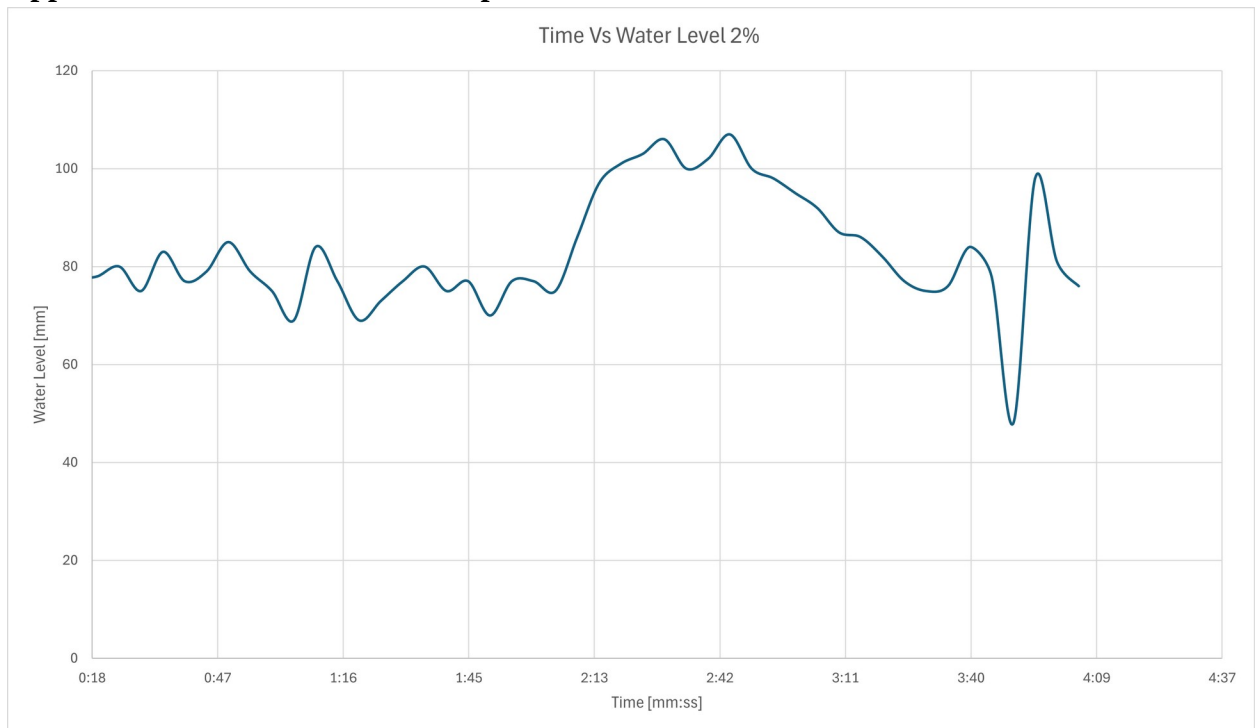
every minute, it is expected that the drain hole of the vessel containing 60g of water would experience a greater pressure than a vessel containing 30g, assuming that both vessels experience the same gravitational constant as described in [5]. And the greater the pressure, the faster the drain rate.

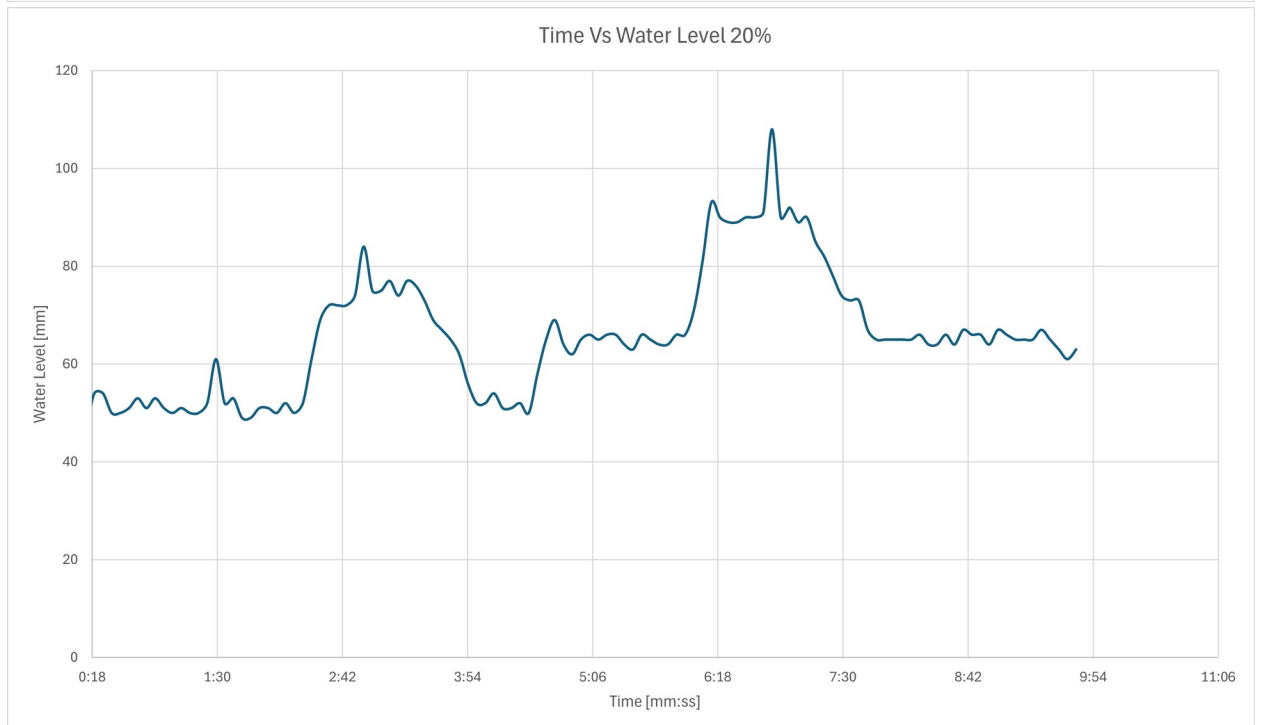
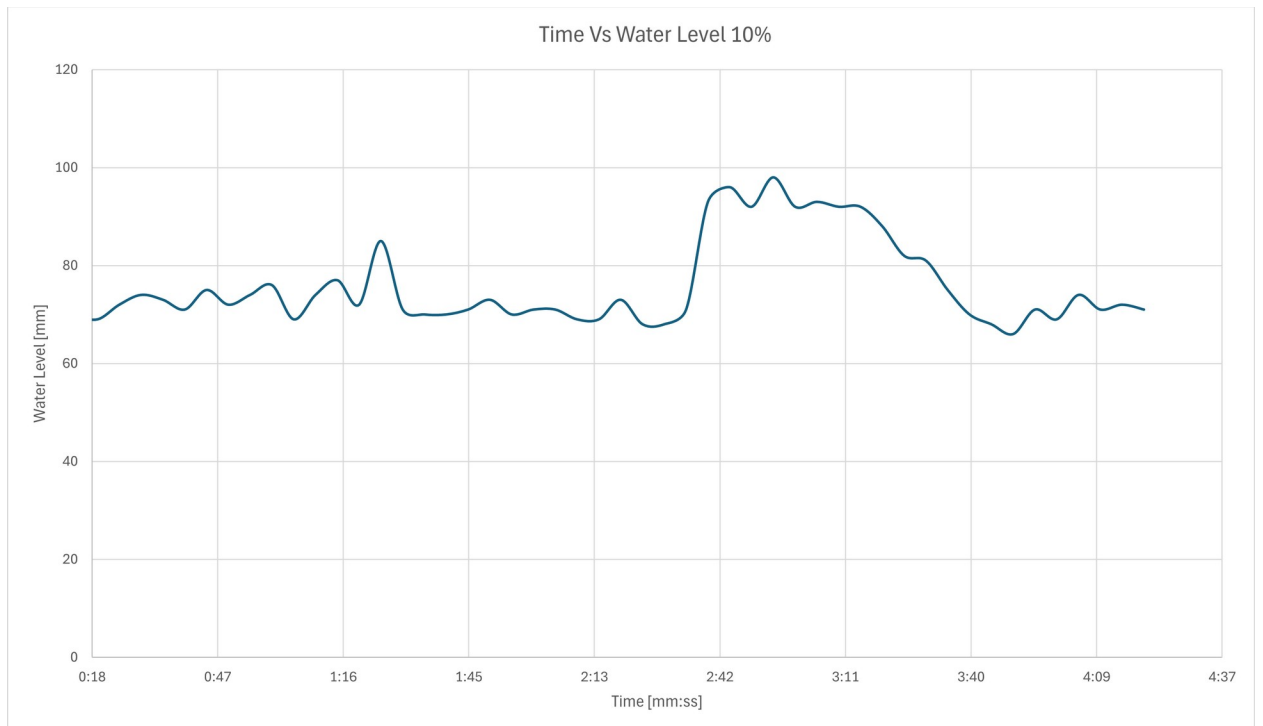
In terms of process control it is ideal to have a P value low enough to respond to disturbances rapidly but not one so low that it becomes overly sensitive and creates instability in the process. Decreasing the band value increases how aggressive the controller is in attempting to reach the set point for any given disturbance. If the action taken by the controller creates instability it can lead to overcorrections leading to constant oscillations in operation. This phenomenon was seen experimentally in experiment 2 with the conclusion being that the ideal band value was 5%. This value can be further improved by implementing integral control as was explored in experiment 3.

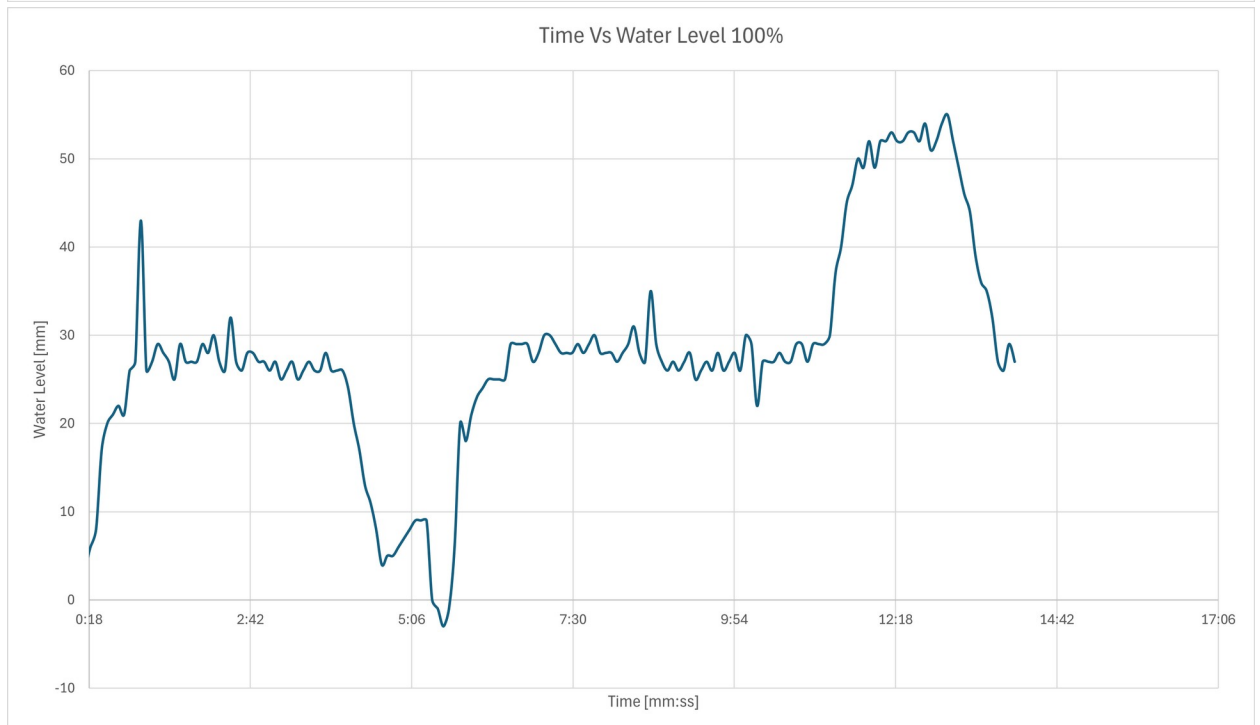
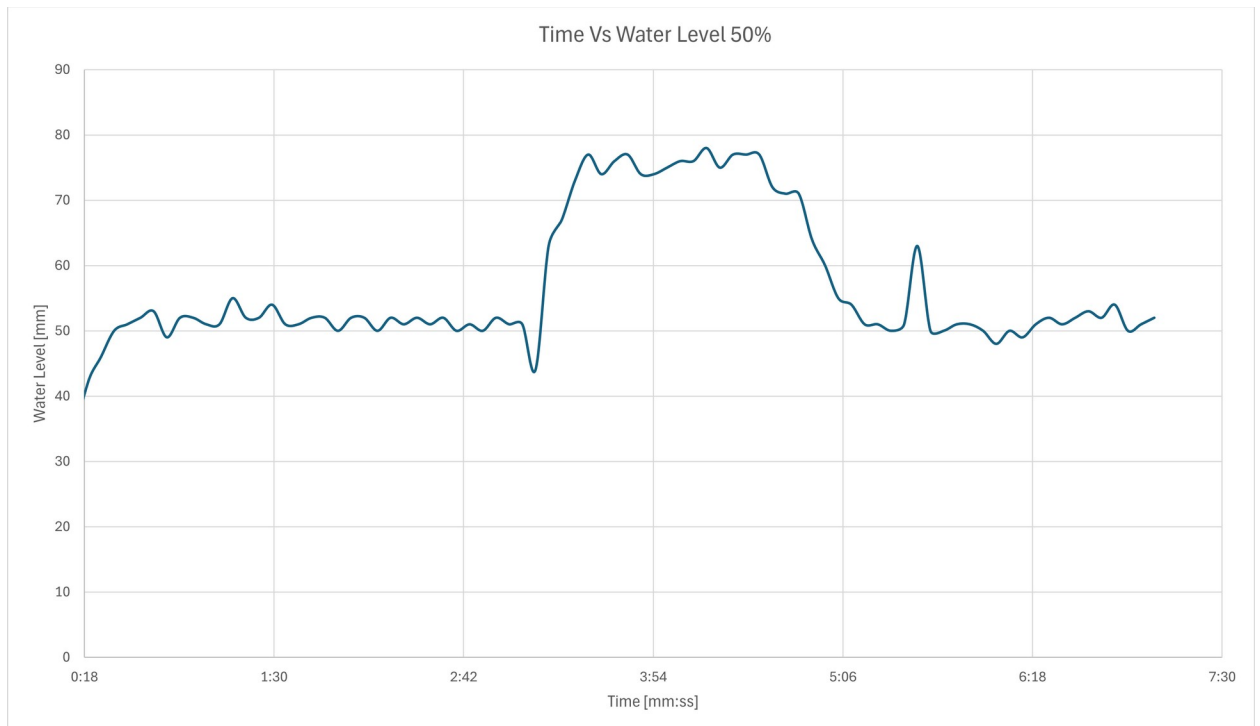
Experiment #3 determined that the optimal integral time was 50s as that is where it had the smallest offset while maintaining stable operation. Although an integral time of 25s was able to strictly maintain level with an open solenoid valve, it increased how much the system oscillated resulting in less steady overall results. In practice, it is more important for the controller to maintain a process variable within an acceptable range rather than at one strict value. Therefore, an integral time of 50s is the ideal time as it provides the lowest offset value while maintaining high stability and acting quickly.

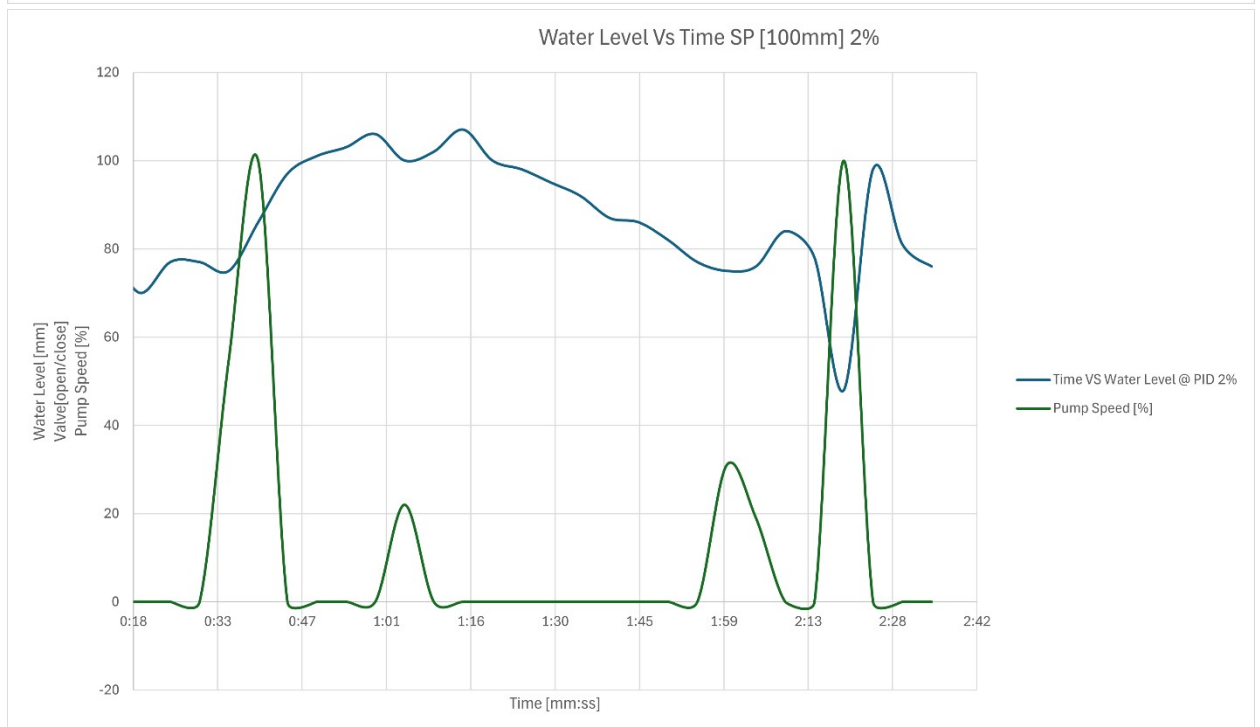
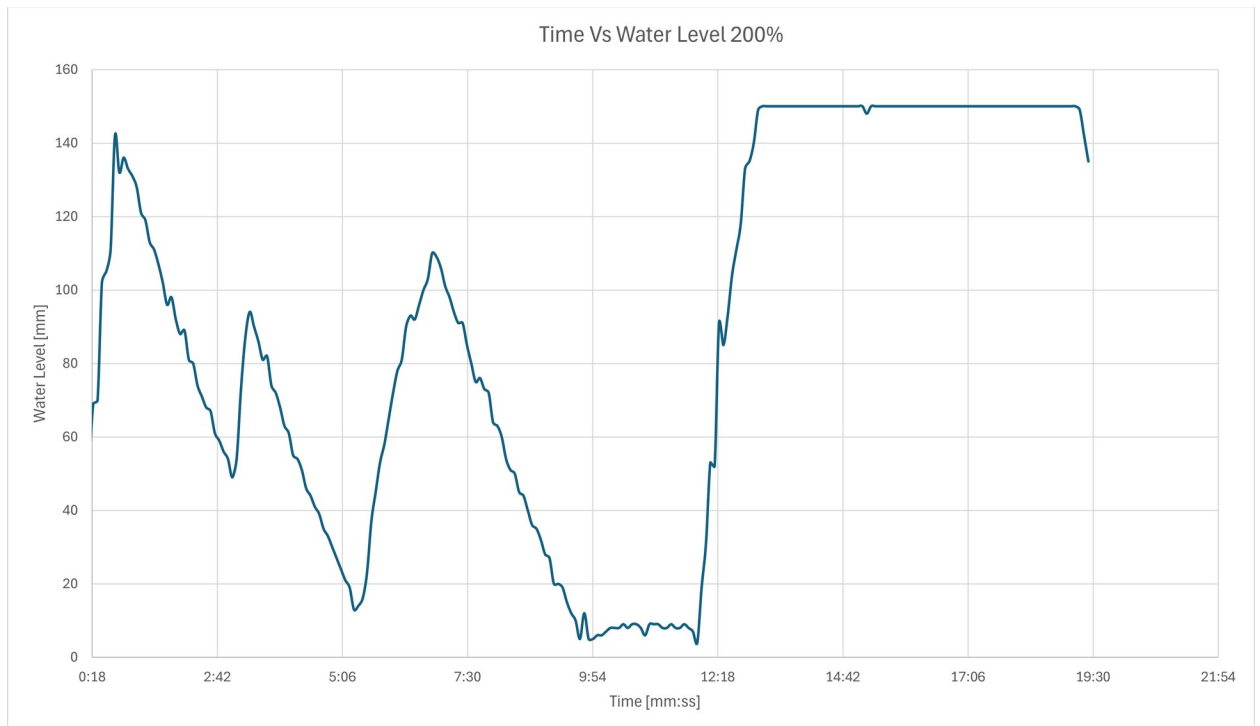
6. **References**

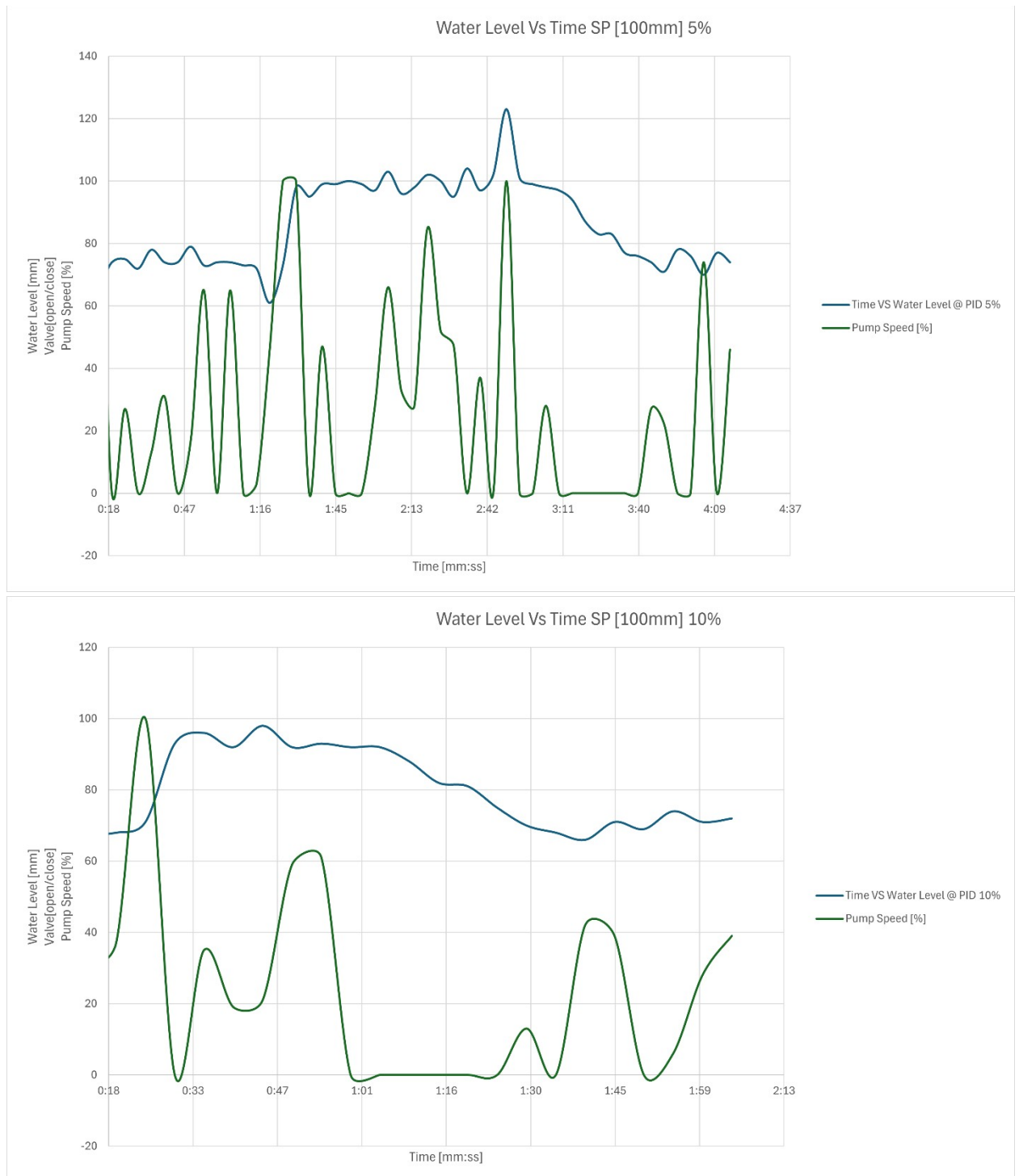
- [1] <https://www.oempanels.com/process-control-applications>
- [2] Process Control Lab Manual
- [3] Perry's Chemical Engineers' Handbook, Eighth Edition
- [4] <https://pubchem.ncbi.nlm.nih.gov/compound/Water>
- [5] <https://physics.nist.gov/cgi-bin/cuu/Value?bg>

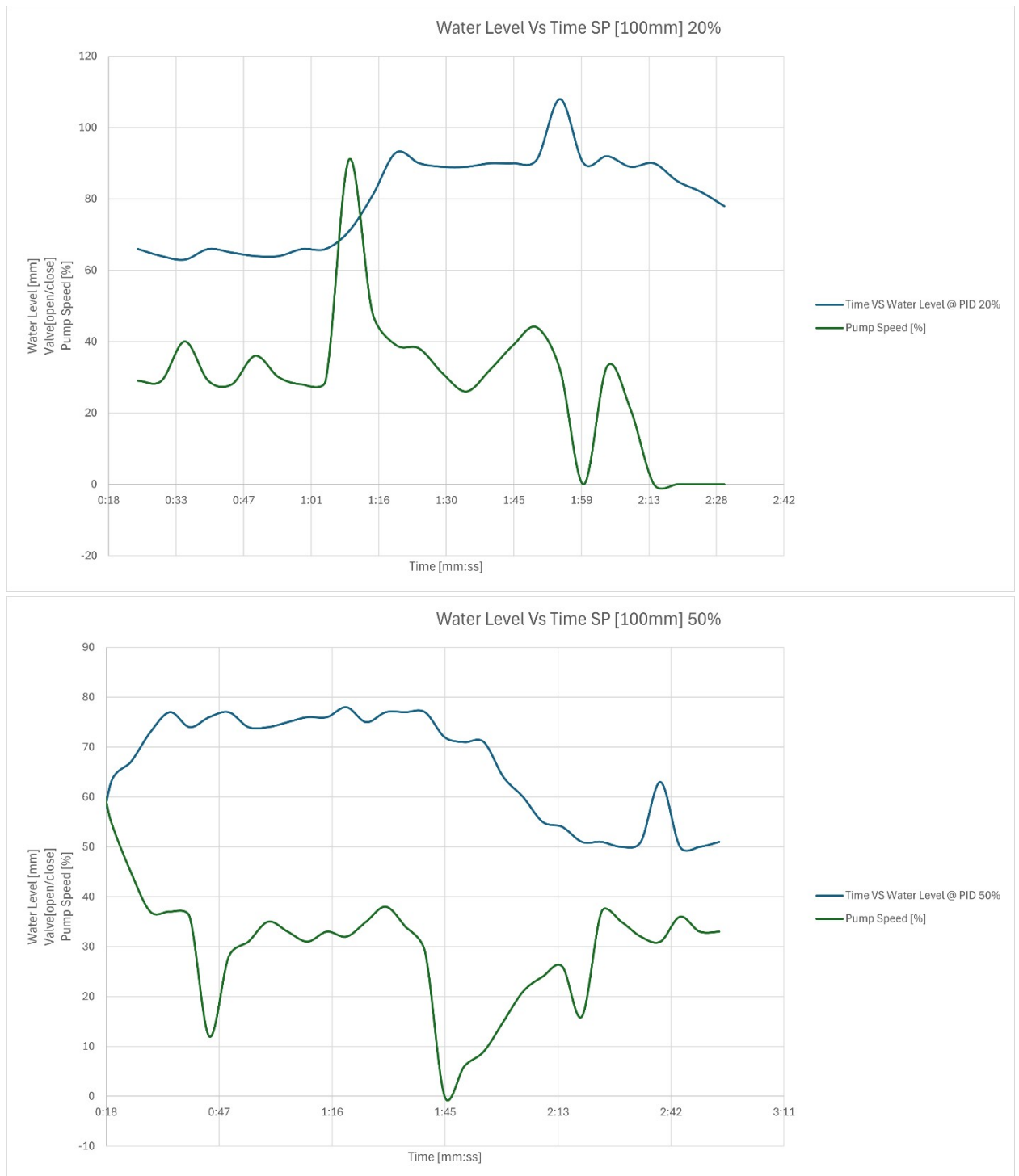
7. Appendix I: Data Tabulations/Graphs

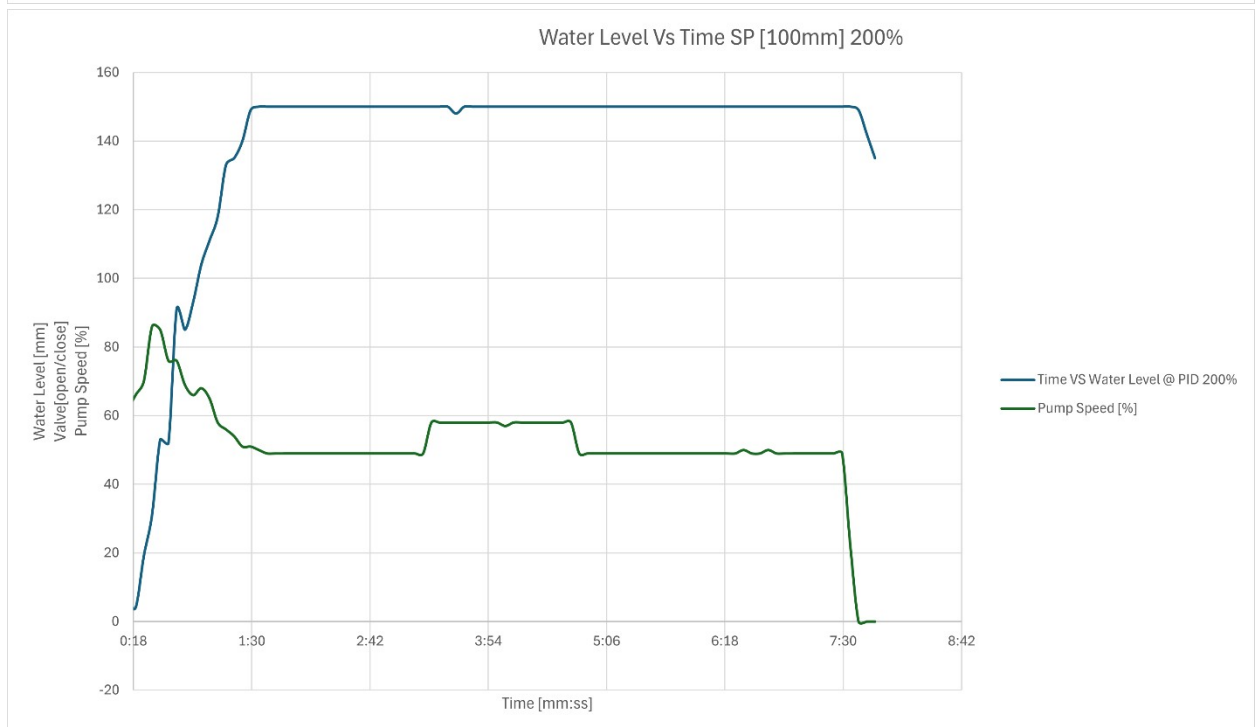
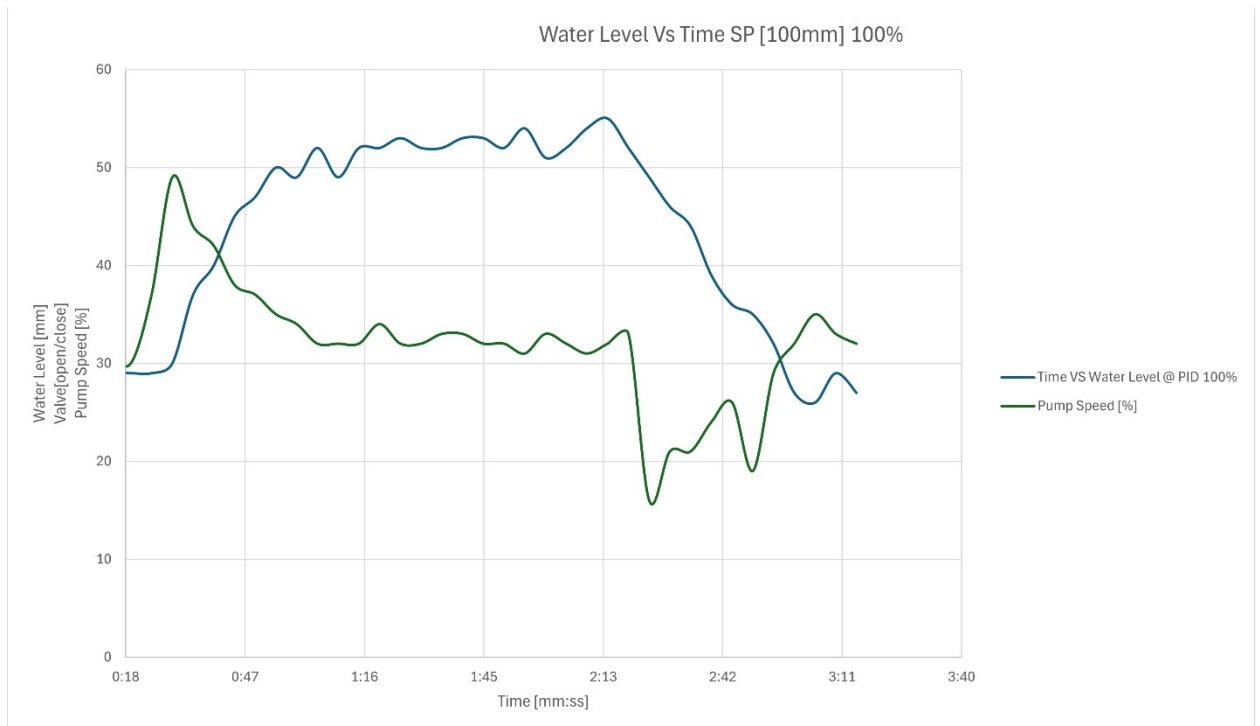


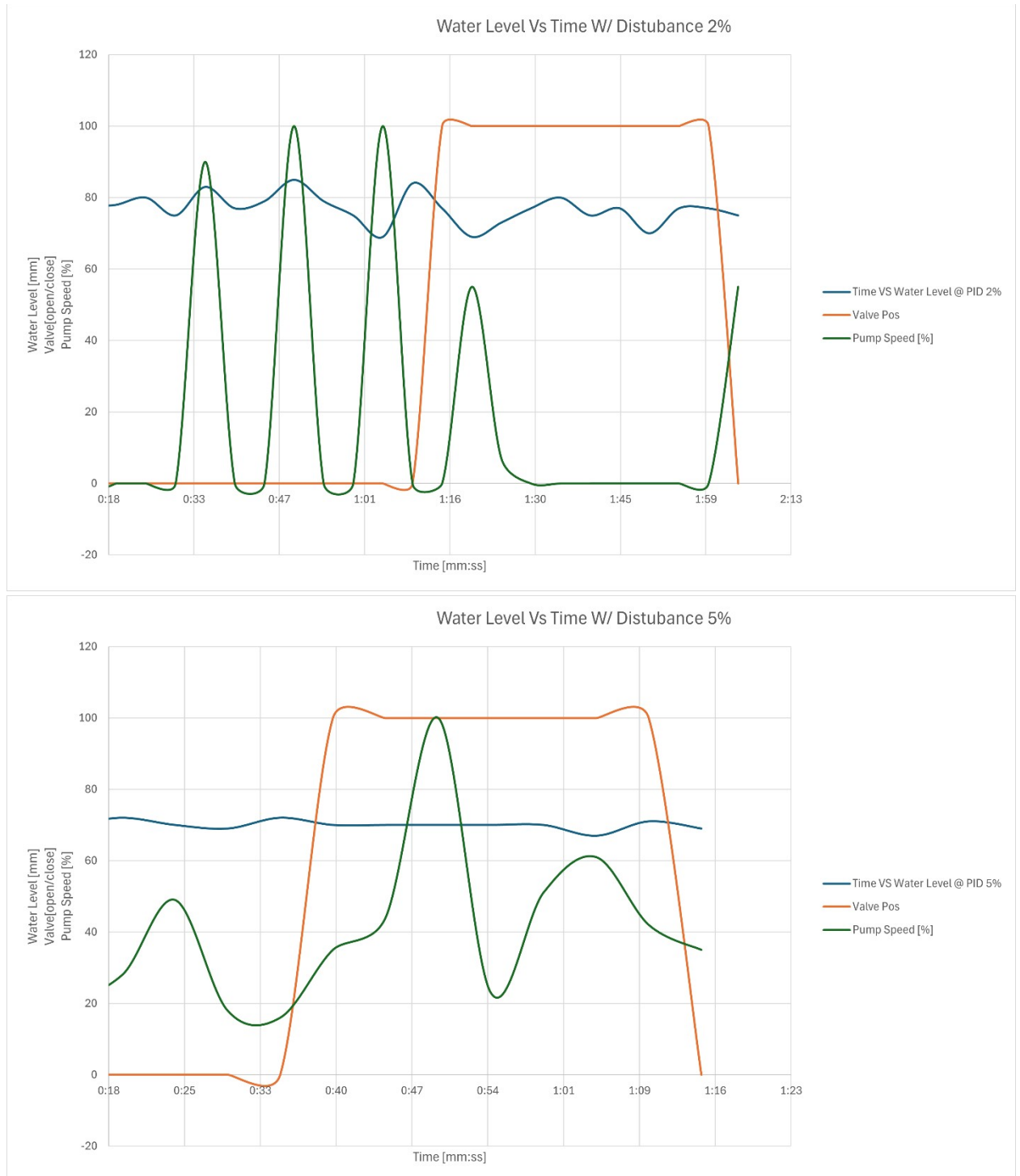


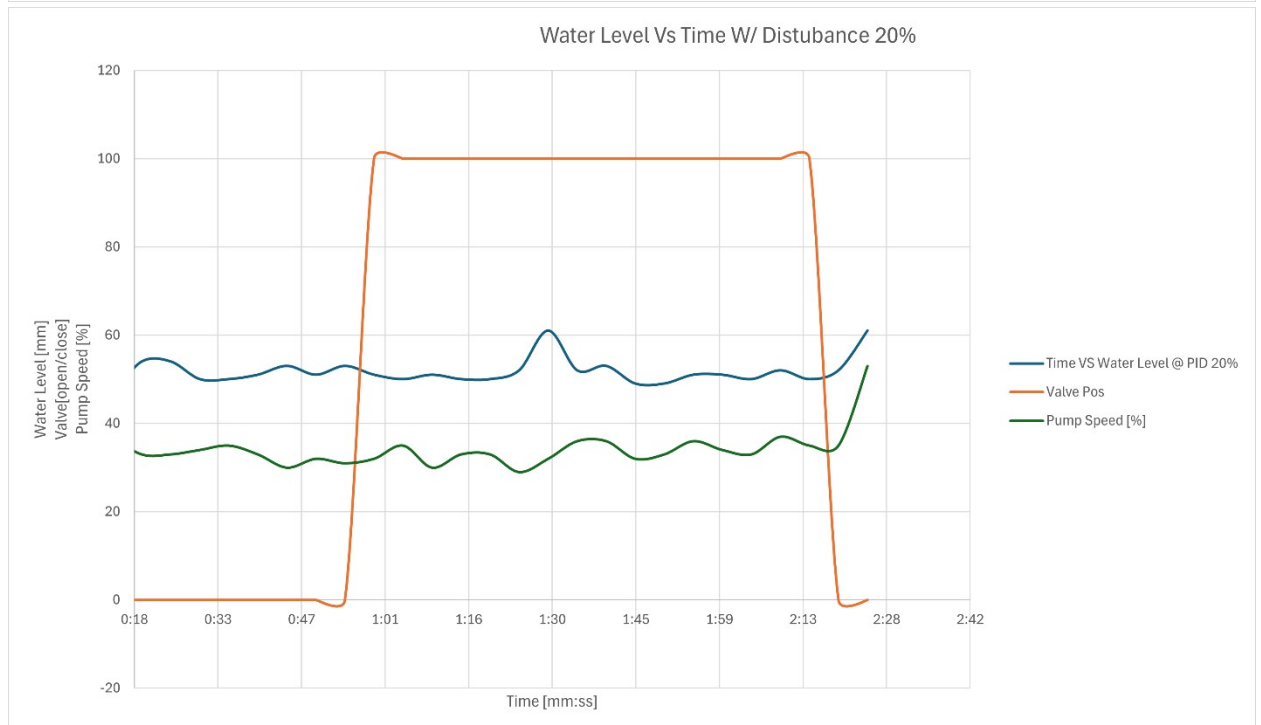
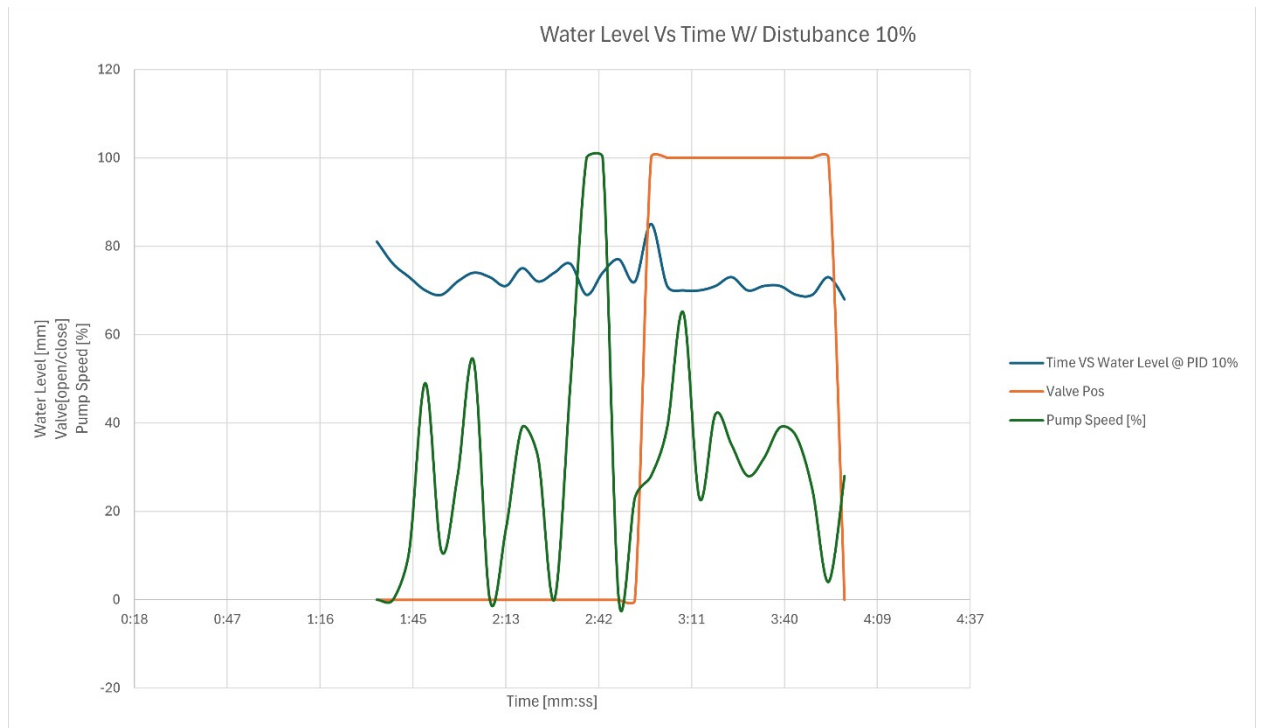


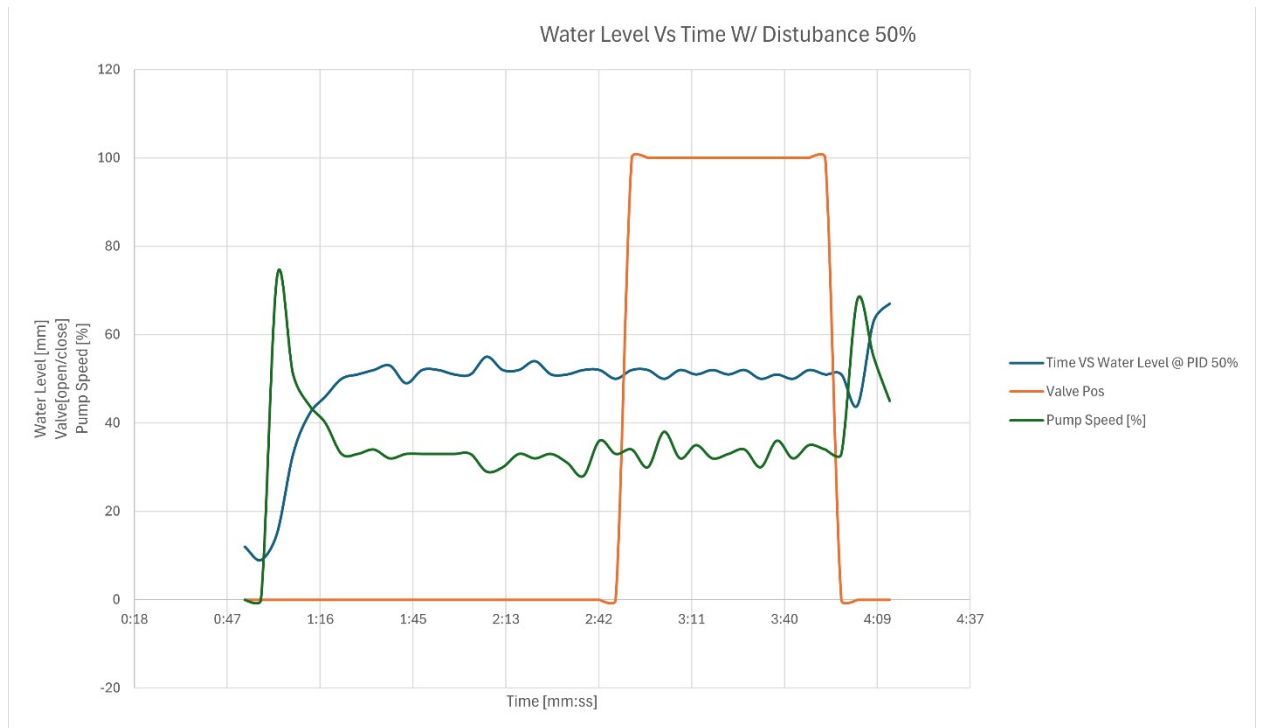


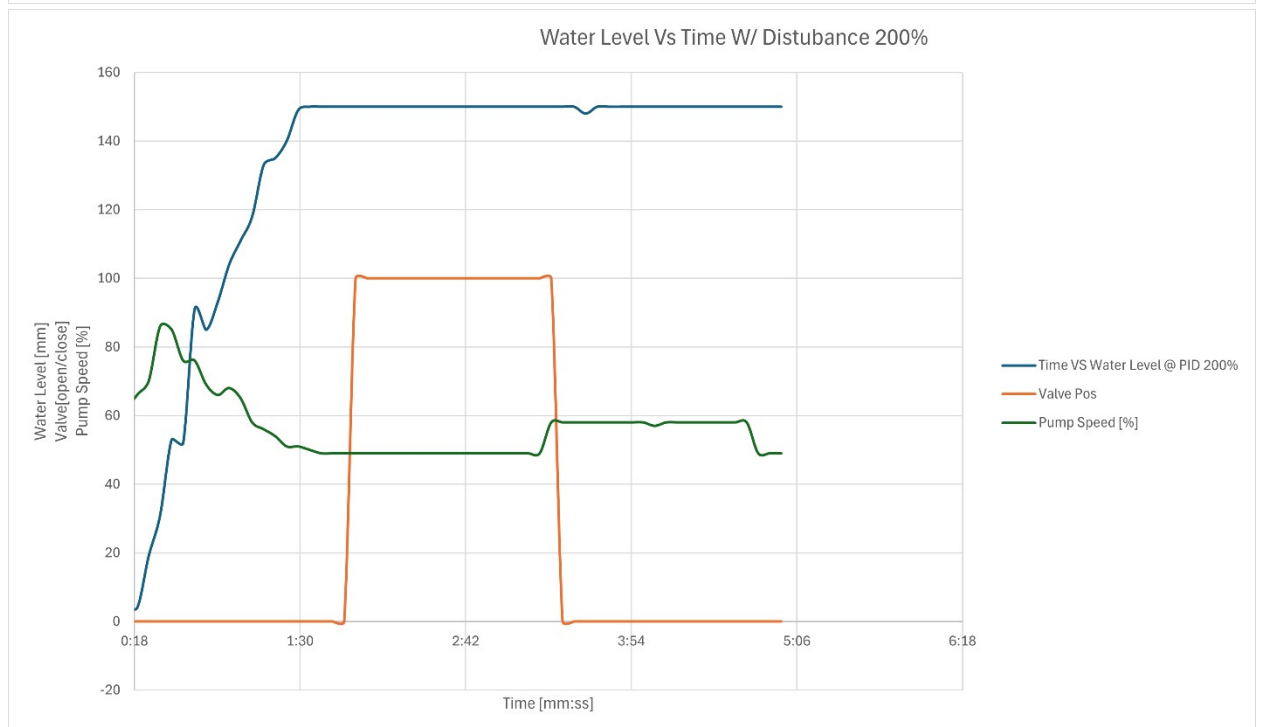
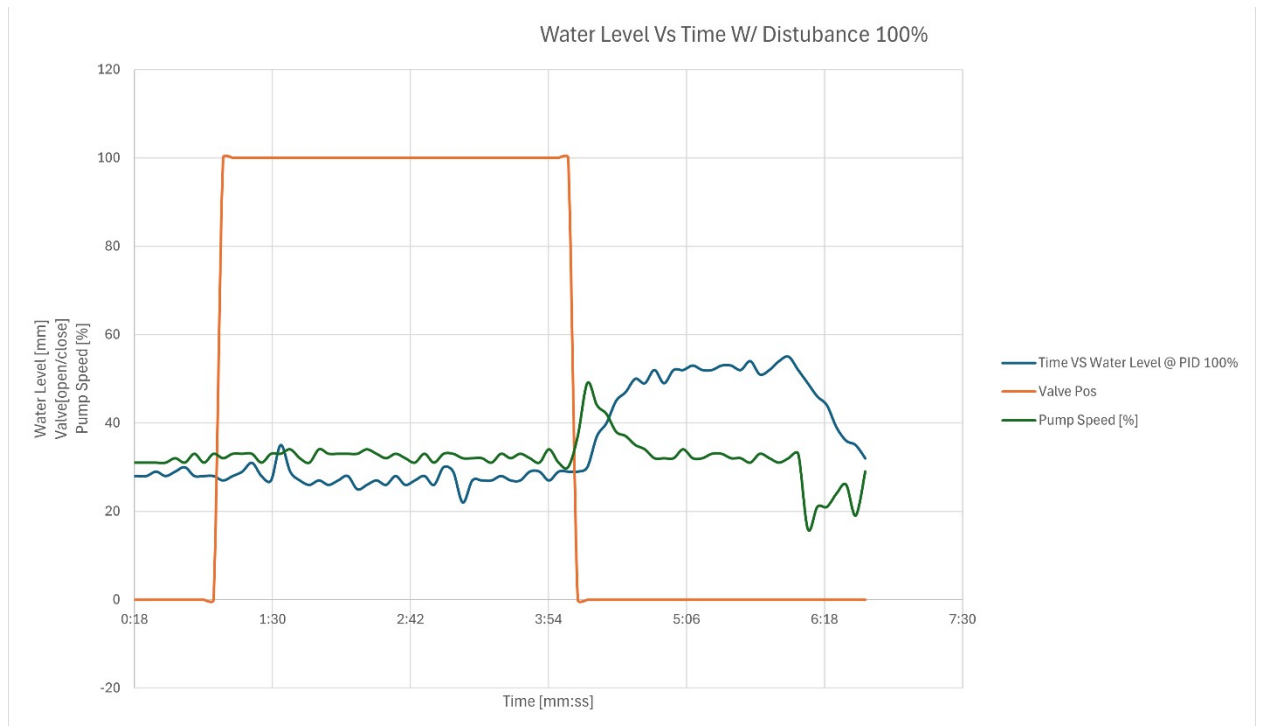




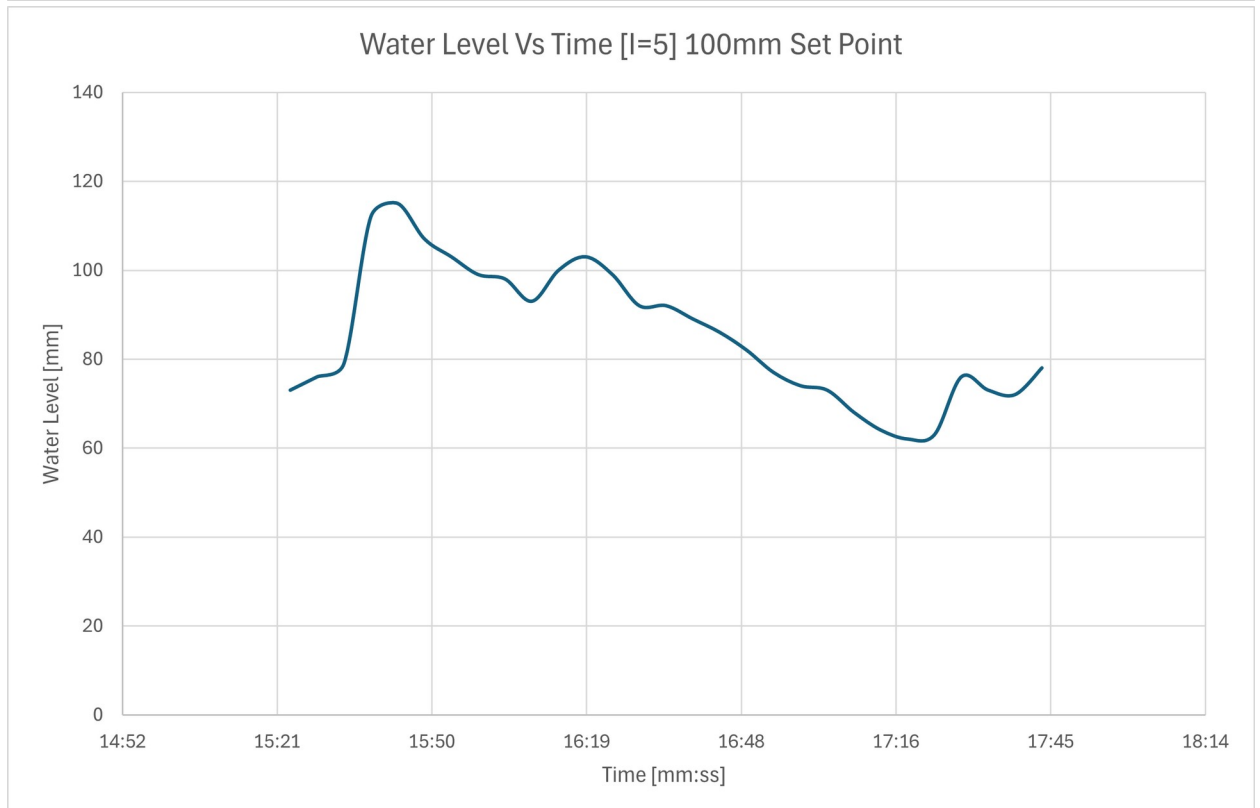
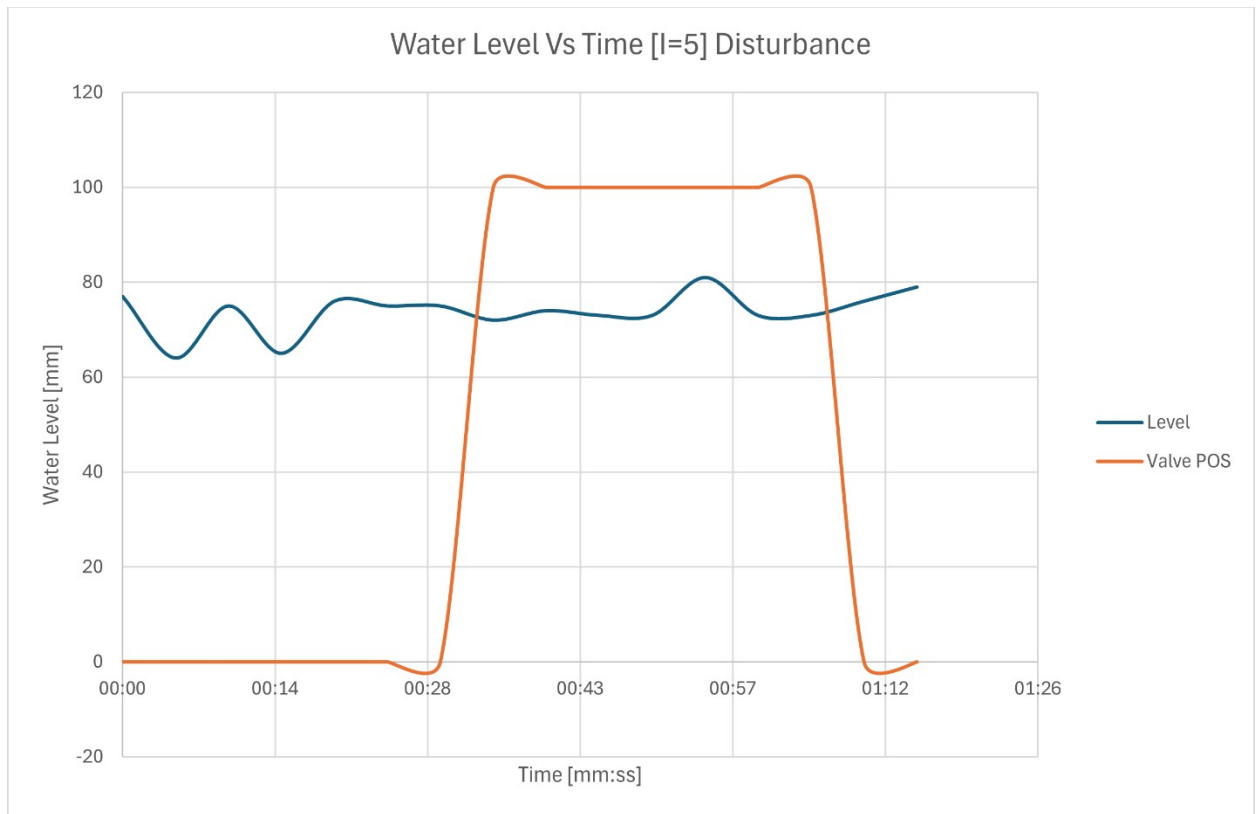


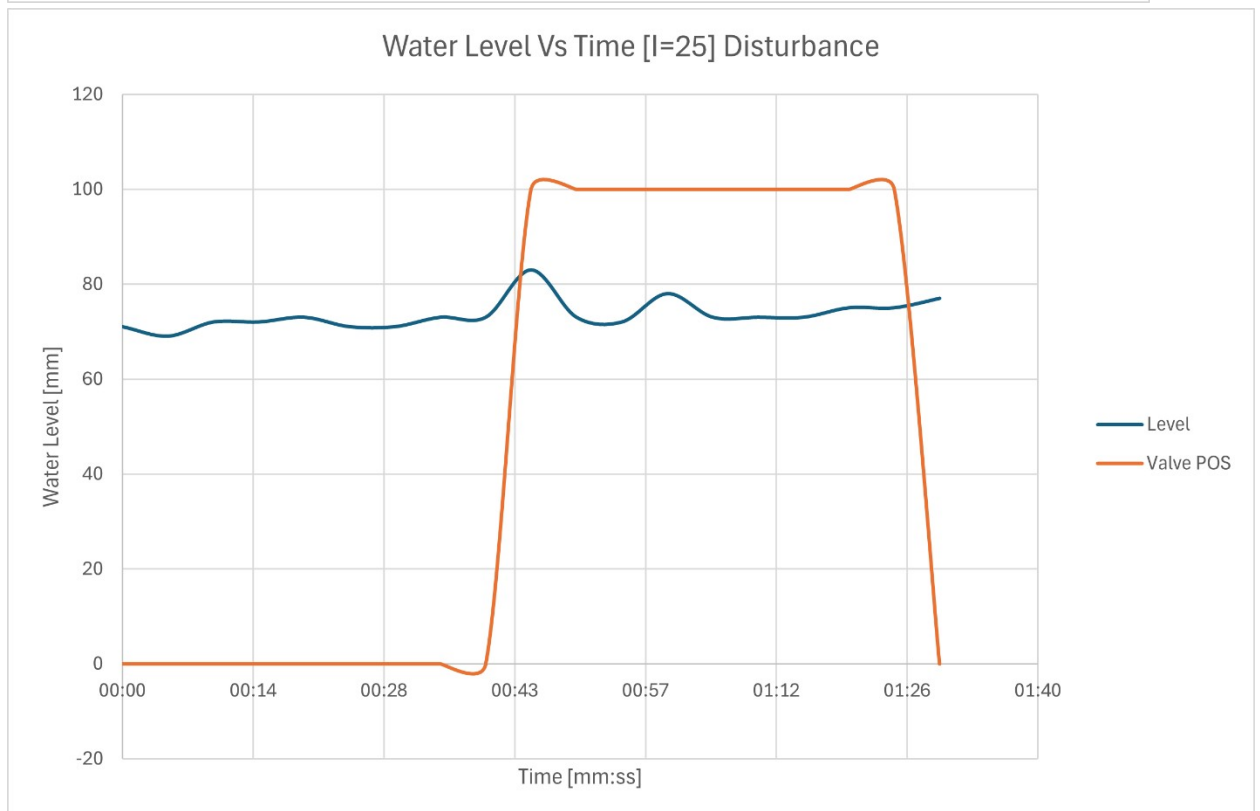
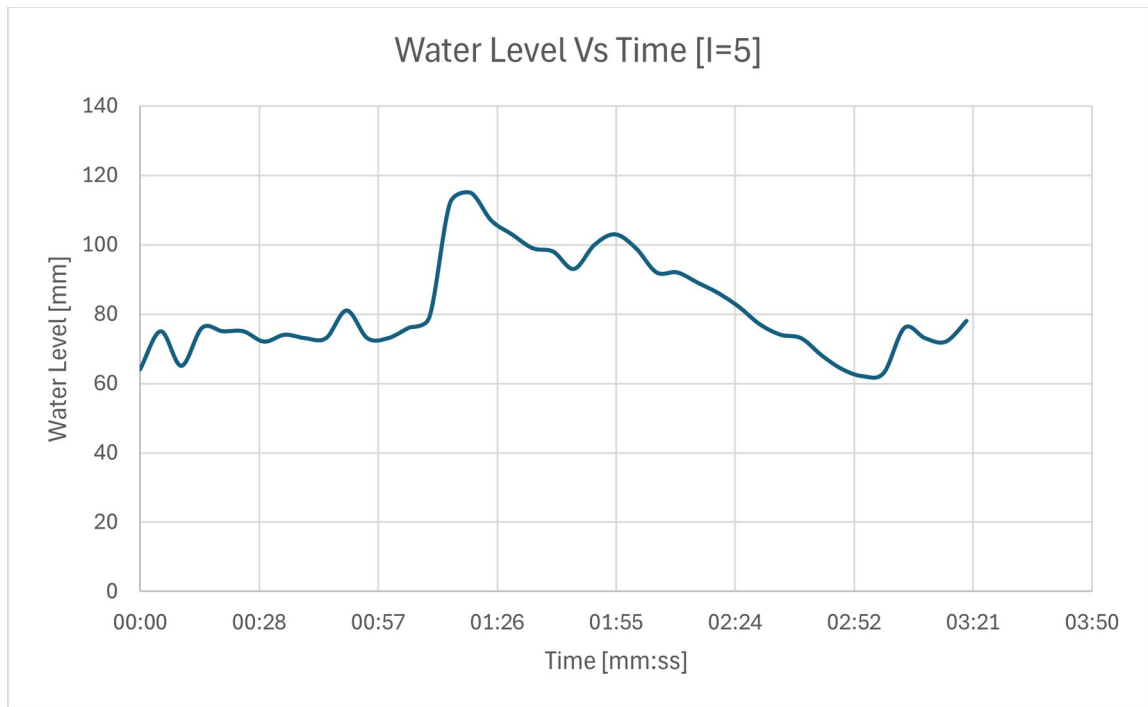


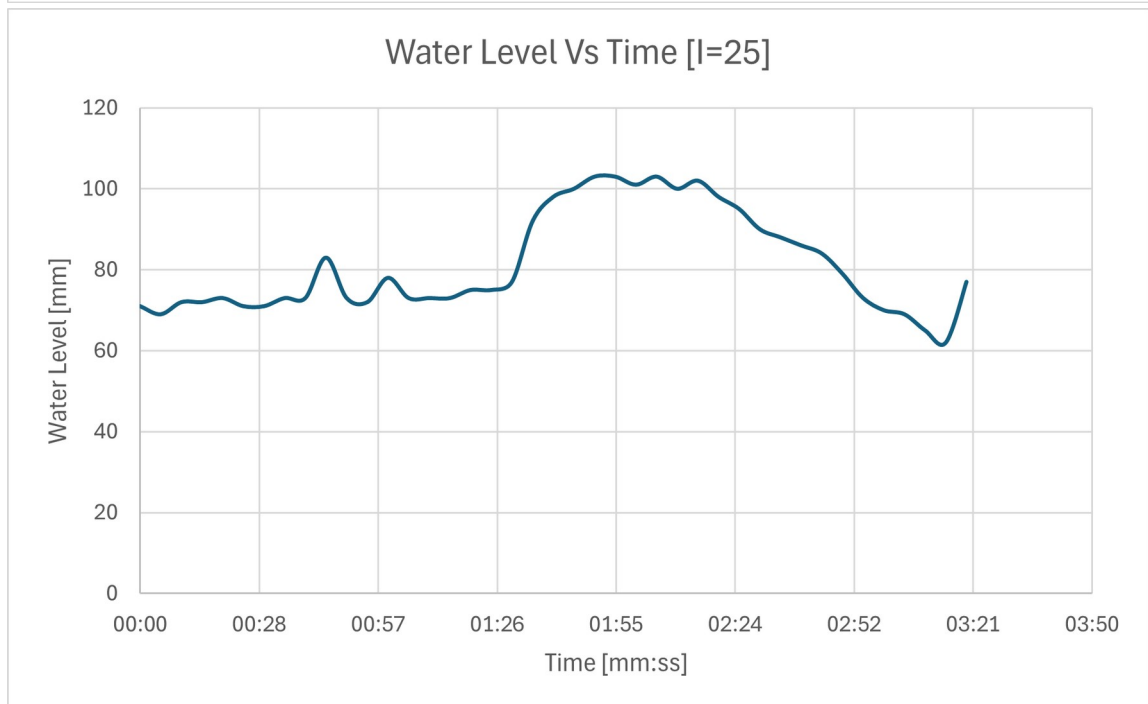
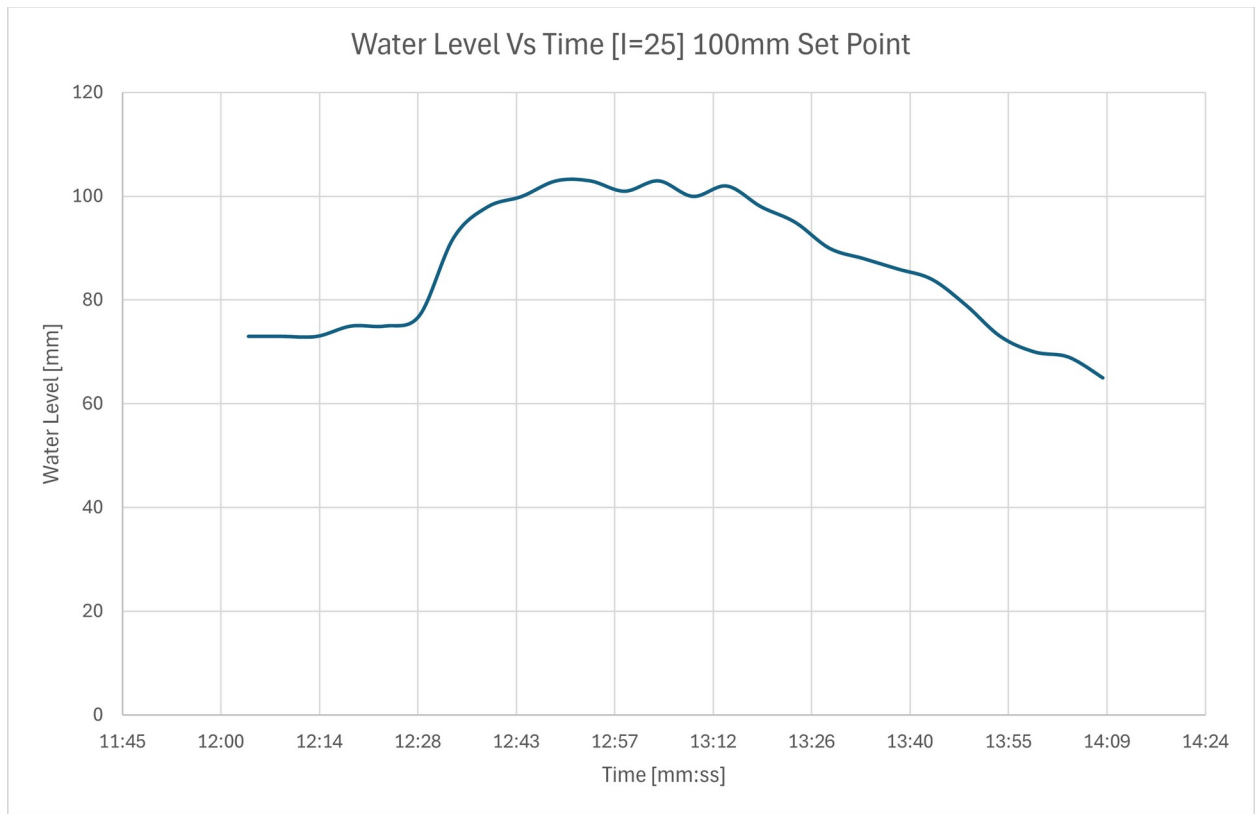


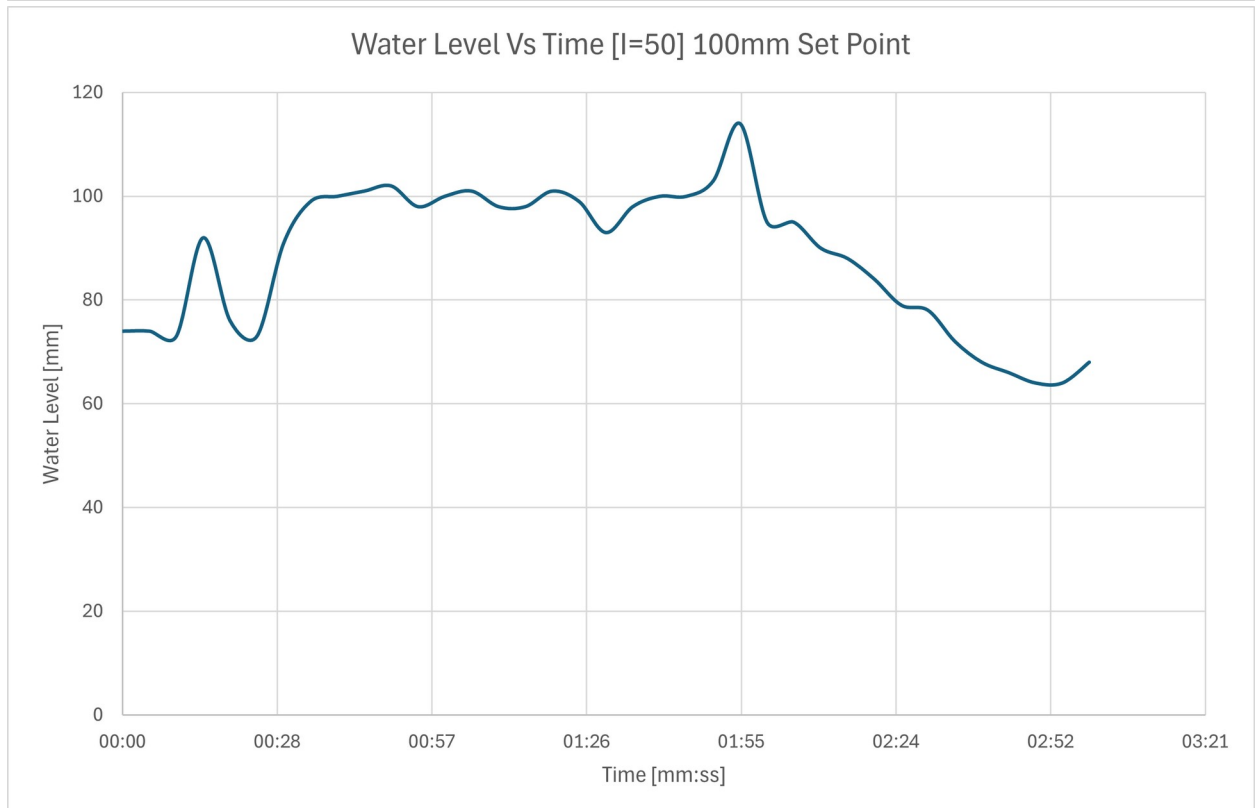
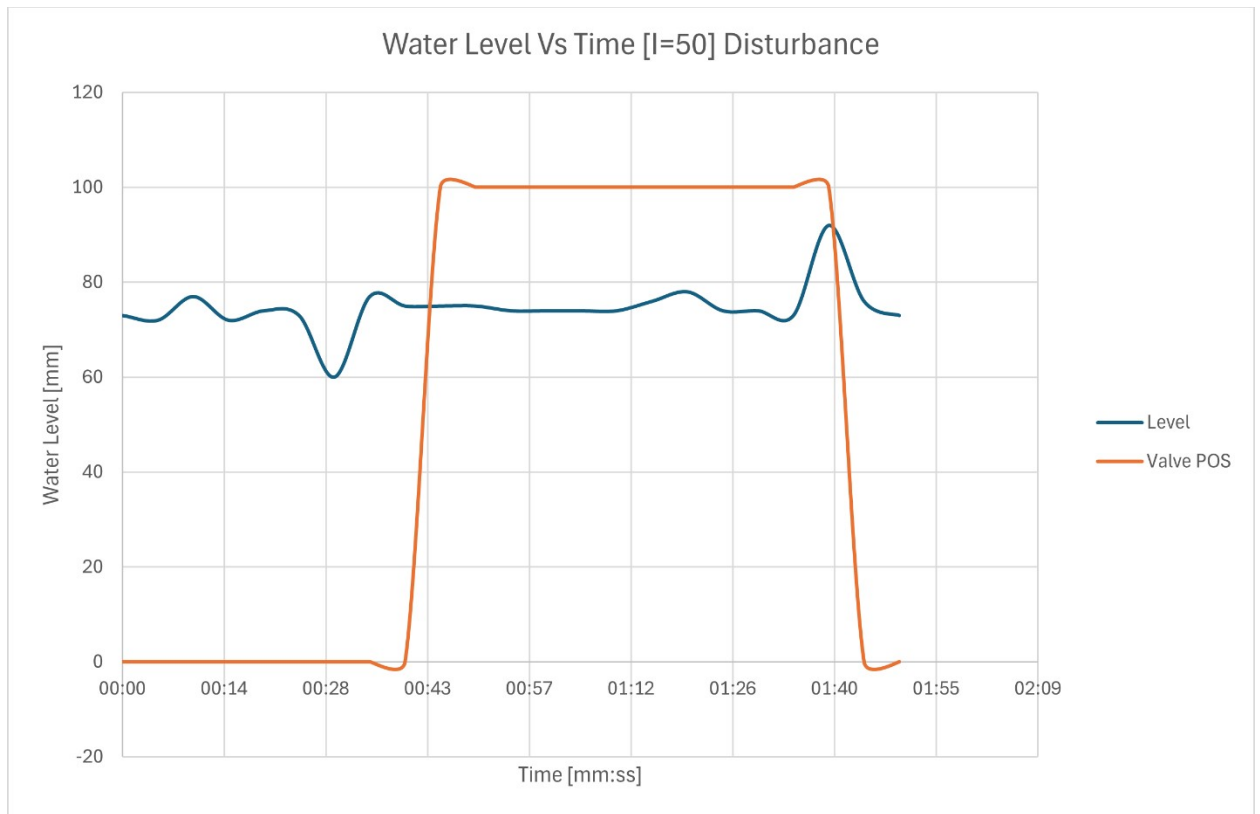


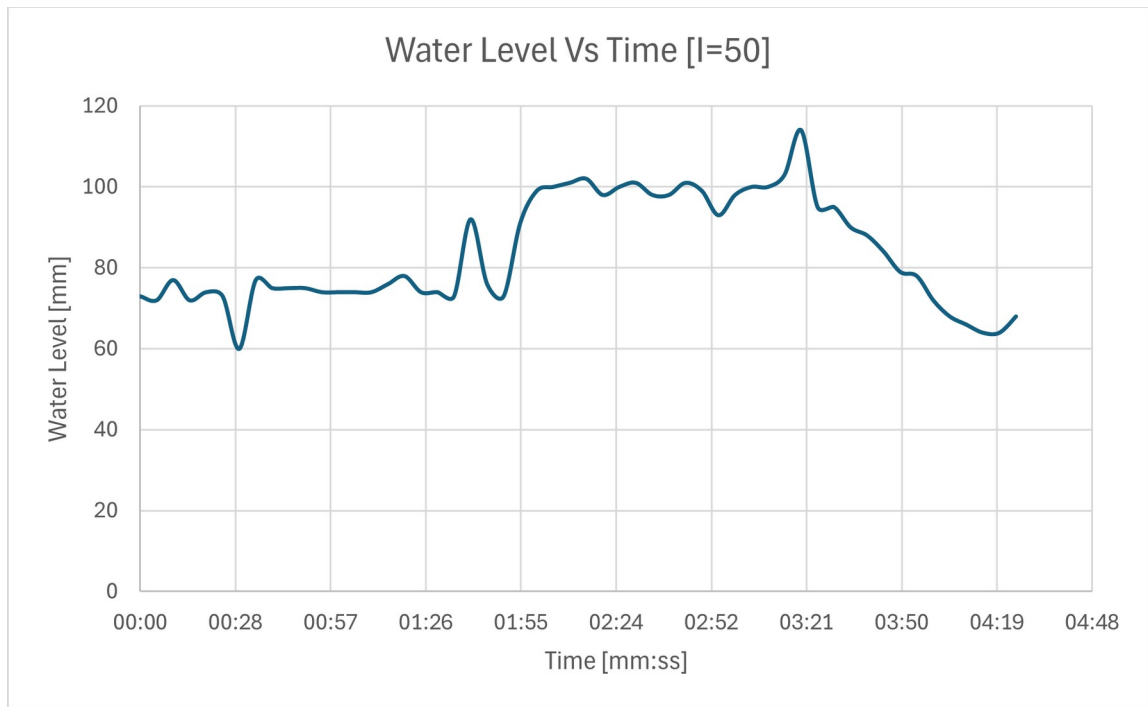
Experiment 3

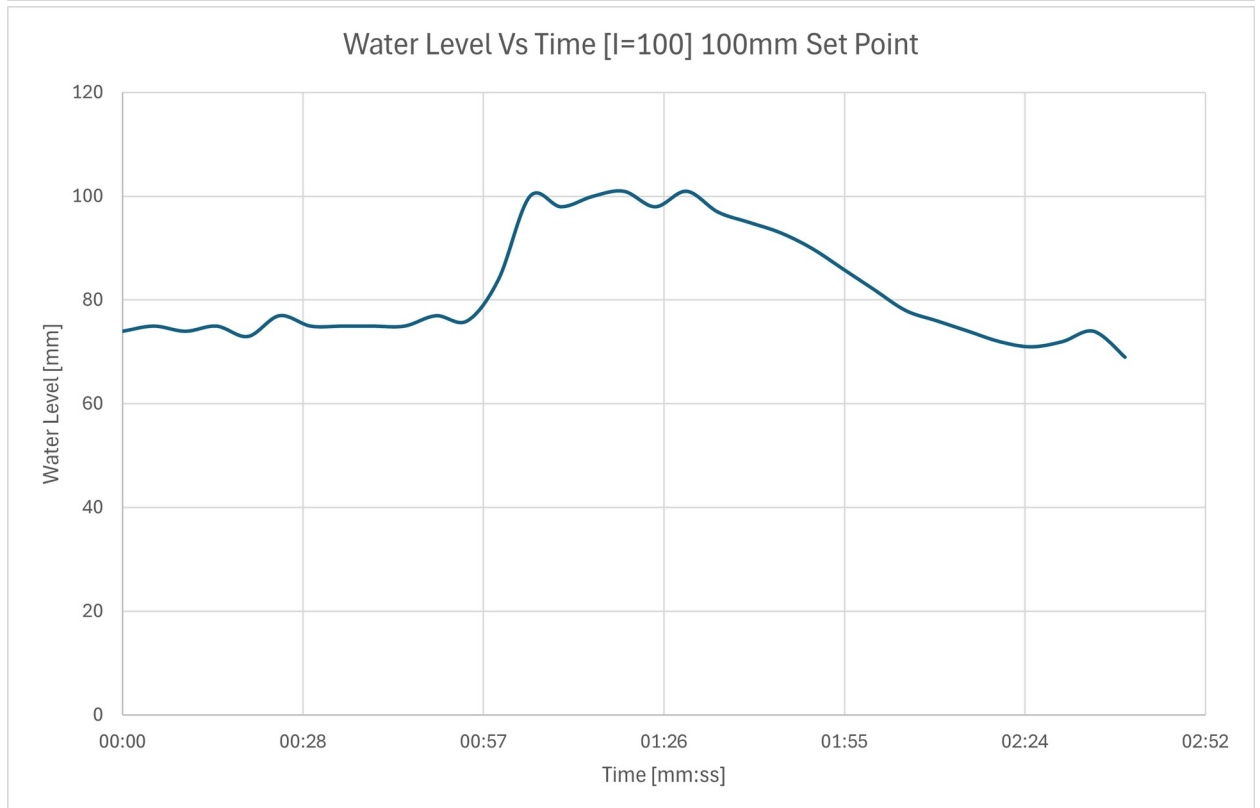
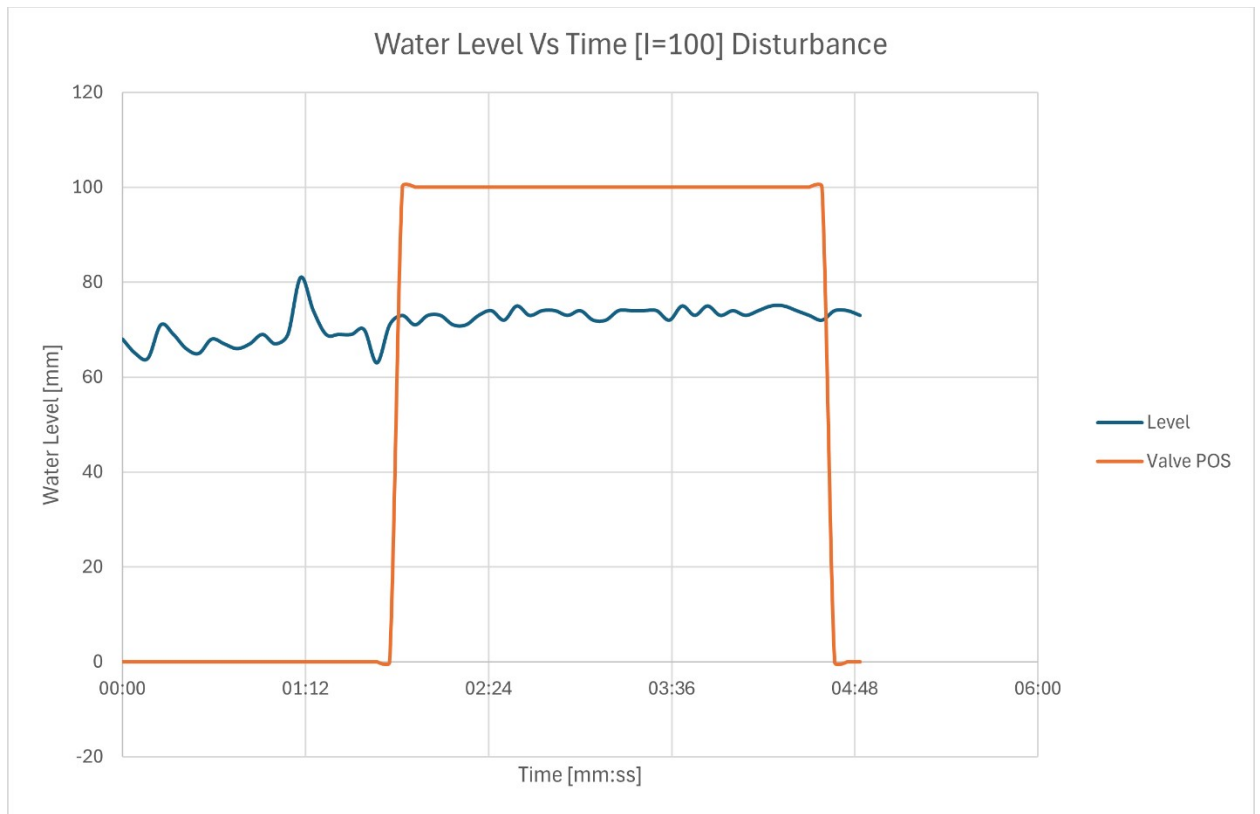


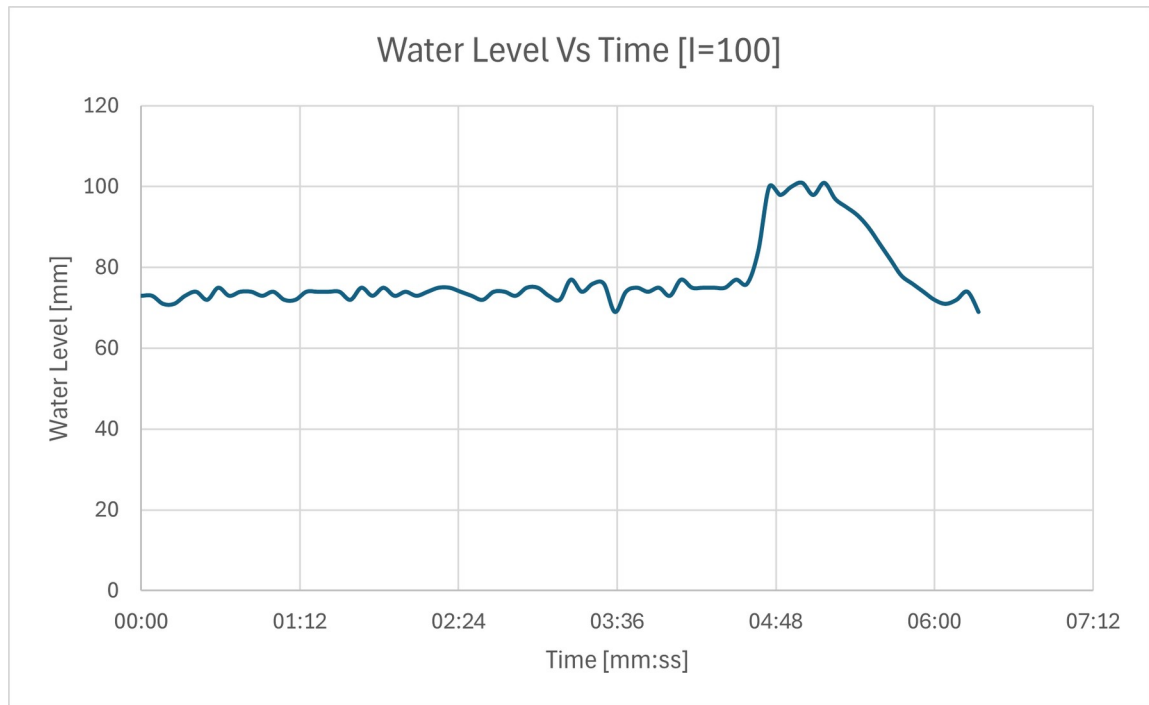












8. Appendix II: Error Analysis

The biggest sources of error in this experiment come from the level sensor and the software that is recording the data. Since the water level is not measured directly, it uses a pressure sensor at the bottom of the upper tank. If the sensor is not perfectly zeroed before starting, all the level readings can be slightly off. This makes the offset that is calculated look larger or smaller than what it really is. During the 3rd experiment there were issues with the level transmitter as it displayed erroneous data. Another source of error comes from how the pump and solenoid behave in the system. The electrical noise from the pump and solenoid can cause small jumps in the level reading. Even when the water isn't changing. The small spikes in the data can make it harder to calculate the offset accordingly and harder to distinguish when the tank has reached steady state. Another source of error comes from the software. Another source of error is the pump speed doesn't translate perfectly to flow rate. The percentage speed can produce slightly different inflows between trials. The solenoid valve being delayed in opening or closing can cause delay in data collection leading to some skewed data.

These sources of error can be reduced by improving how the system is prepped and how the data is analyzed. Level sensors should be zeroed with the vessel empty before each run and checked against the tank scale at a known height to prevent any shifts in the readings. Electrical noise can be reduced by keeping the sensor cables separate from the pump and solenoid power wires. To make the pump more response consistent the inlet valve position should remain the same throughout all the trials.

9. Appendix III: Sample Calculations

Only a small amount of calculations were required for this experiment. Most of the numerical work was done using Excel formulas to calculate the average offset values from the collected data.

10. Appendix IV: Individual Team Contributions

ASSIGNMENT	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	All Team members
Pre-Lab Editing	2	All Team members
Experimental Objective (Prelab)	1	Rishabh Patel
Apparatus (prelab)	1	Khoa Nguyen & Miguel Piza
Materials & Supplies (prelab)	1	Khoa Nguyen
Procedure (prelab)	1	Alexis Guillen
Project Safety Evaluation (prelab)	1	Kyle Hankosky
Final Lab Editing	1	All Team members

Executive Summary (final lab)	2	Khoa Nguyen & Rishabh Patel
Introduction (Final lab)	1	Khoa Nguyen & Rishabh Patel
Theory (Final Lab)	4	Rishabh Patel & Khoa Nguyen
Results (final lab)	2	Kyle Hankosky & Alexis Guillen & Miguel Piza
Discussion (final lab)	2	Khoa Nguyen & Miguel Piza
References (final and/or prelab)	1	Rishabh Patel
Data Tabulation/Graphs (final)	2	Kyle Hankosky & Alexis Guillen & Miguel Piza
Error Analysis (final lab)	1	Kyle Hankosky & Alexis Guillen
Sample Calculations (final Lab)	2	Kyle Hankosky & Alexis Guillen
PowerPoint Presentation	5	All Team members
Total	35 Hours	